

TECHNOLOGY



Designing Microsoft Azure Infrastructure Solutions: AZ:305

Design Authentication Solutions



Learning Objectives

By the end of this lesson, you will be able to:

- Describe Microsoft Entra ID to effectively manage and streamline user authentication
- Evaluate and propose an appropriate solution for single sign-on (SSO) to enhance user authentication efficiency and security
- Formulate a conditional access strategy to control user access based on specific conditions
- Design and justify the implementation of a solution incorporating managed identities to enhance security and simplify identity management processes



Learning Objectives

By the end of this lesson, you will be able to:

- 🕒 Device a solution integrating Key Vault for secure storage and management of sensitive information
- 🕒 Develop a hybrid identity solution that seamlessly integrates on-premise and cloud-based identities
- 🕒 Recommend a solution for B2B integration, fostering secure collaboration and data exchange between external partners and organizations

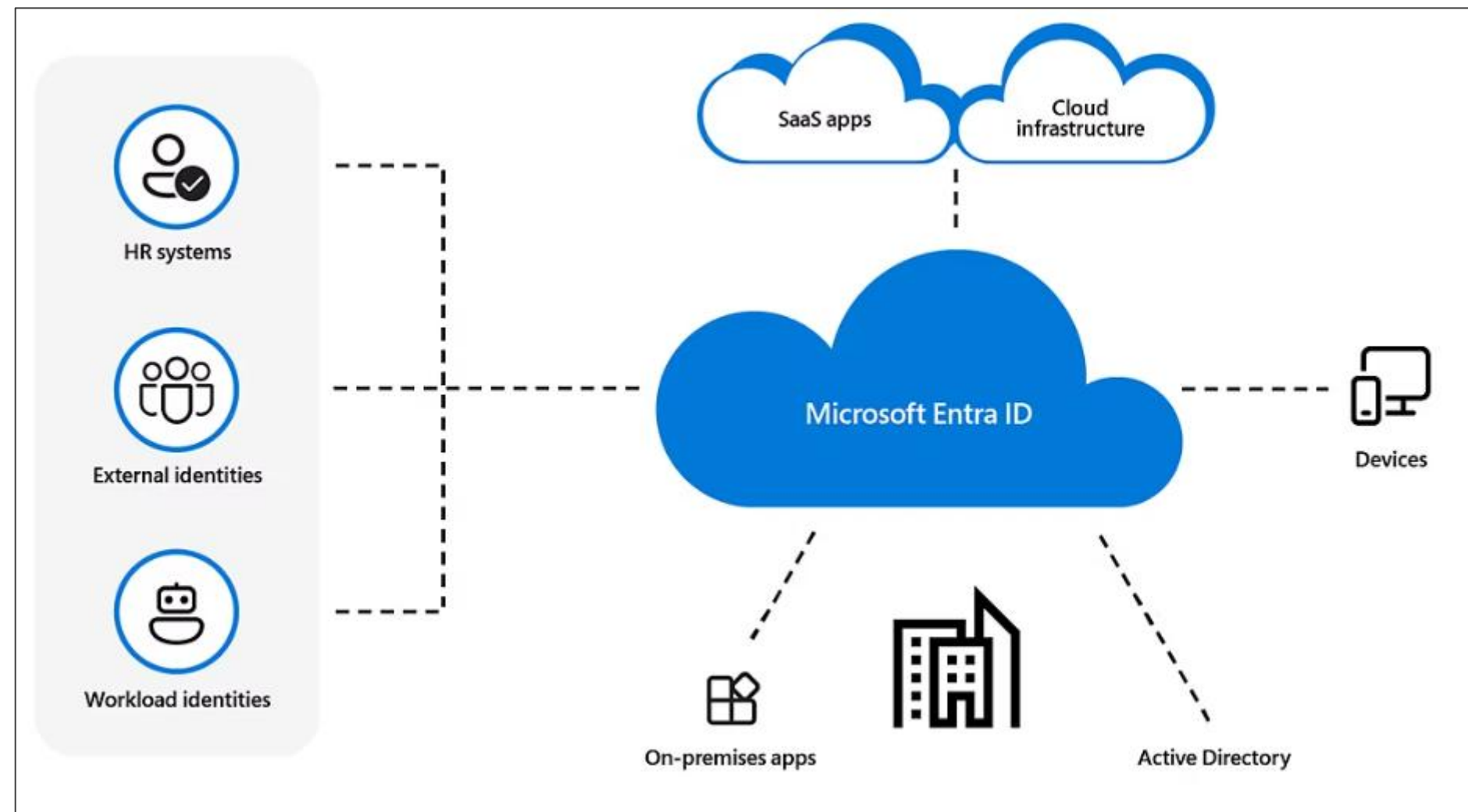


TECHNOLOGY

Microsoft Entra ID

Microsoft Entra ID

It is a cloud-based identity and access management service, which helps access external resources.



Source: <https://www.microsoft.com/en-in/security/business/identity-access/microsoft-entra-id#diagram-cta-popup>

Microsoft Entra ID: Features

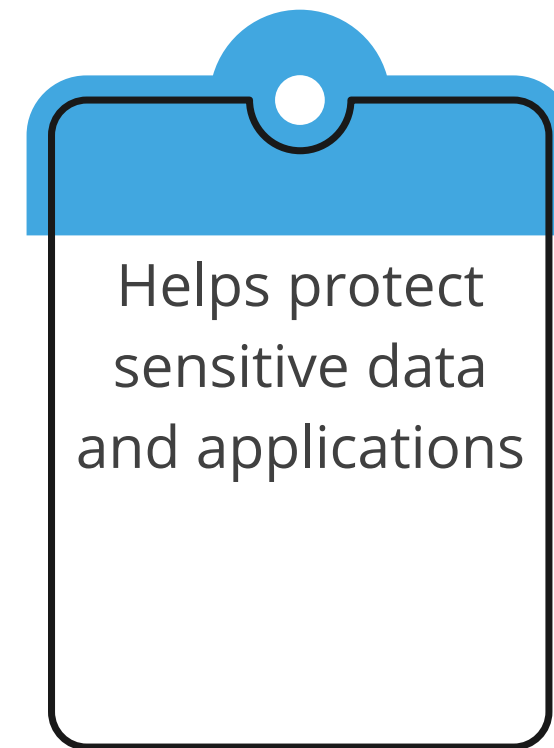
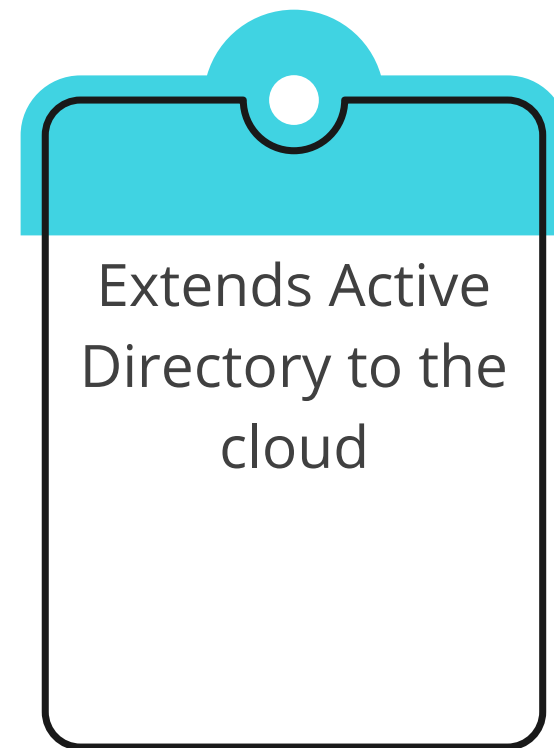
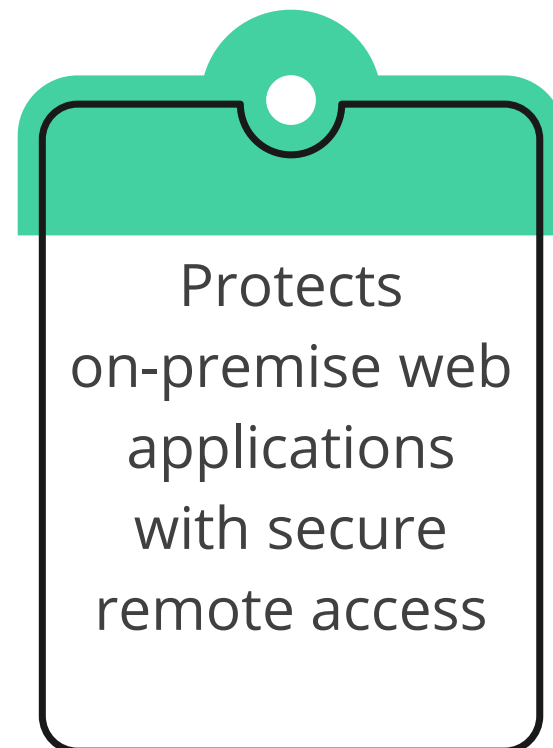
Single sign-on
access to thousands
of cloud
applications and
resources

Identity
management
capabilities and
integration

Integration with
Windows Server
Active Directory

Facilitate the
development of
applications with a
global scope

Microsoft Entra ID: Benefits



Microsoft Entra ID: Benefits

Some additional benefits include:

B2B
collaboration

Dynamic
groups

Group-based
licensing

Addition of
custom
cloud apps

Identity
protection



Microsoft Entra ID: Concepts

Concepts	Description
Identity	It is an object that can be authenticated.
Account	It is an identity that has data associated with it.
Microsoft Entra account	It is an identity created through Microsoft Entra ID or another Microsoft cloud service.
Azure tenant	It is a dedicated and trusted instance of Microsoft Entra ID. It is automatically created when an organization signs up for a Microsoft cloud service subscription.
Microsoft Entra directory	Each Azure tenant has a dedicated and trusted Microsoft Entra directory.
Azure subscription	It is used to pay for Azure cloud services.

Microsoft Entra ID vs. AD Domain Services

Microsoft Entra ID

- Primarily an identity solution, designed for HTTP and HTTPS communications
- Queried using the REST API over HTTP and HTTPS instead of Lightweight Directory Access Protocol (LDAP)
- Includes federation services and many third-party services (such as Facebook)

AD Domain Services (AD DS)

- Core functions responsible for managing users and computers
- Allows system administrators to organize data into logical hierarchies
- Includes security certificates, single sign-on, LDAP, and rights management

Microsoft Entra ID

Microsoft Entra ID Free

It is designed to introduce system administrators to Microsoft Entra ID.

Microsoft Entra ID P2

It has all the features of the other editions, including advanced identity protection and privileged identity management.



Microsoft Entra ID Governance

It is an advanced set of identity governance capabilities for Microsoft Entra ID P1 and P2 customers.

Microsoft Entra ID P1

It is designed to help organizations with demanding identity and access management needs.

Microsoft Entra ID: Editions

Feature	Free	Microsoft 365 Apps	Premium P1	Premium P2
Directory objects	500,000 objects	No object limit	No object limit	No object limit
Single sign-on	Unlimited	Unlimited	Unlimited	Unlimited
Core identity and access	X	X	X	X
B2B collaboration	X	X	X	X
Identity & access for O365		X	X	X
Premium features			X	X
Hybrid identities			X	X
Advanced group access			X	X
Conditional access			X	X
Identity protection				X
Identity governance				X

Solution for Single Sign-On (SSO)

Microsoft Entra ID Seamless Single Sign-On

It automatically signs in users when they are on their corporate devices connected to a corporate network.

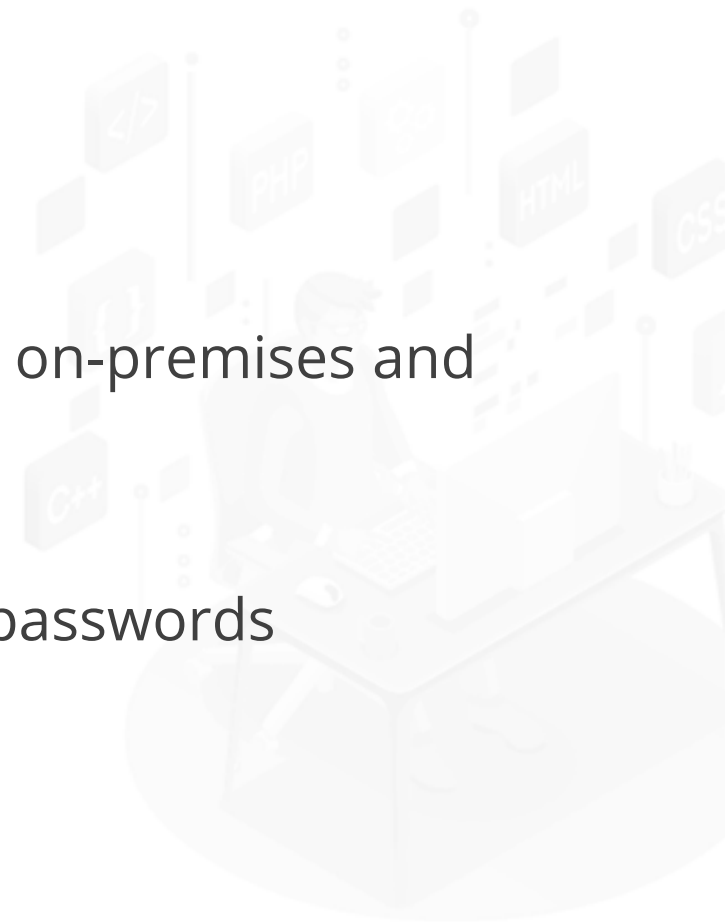


Single Sign-On: Benefits

User experience

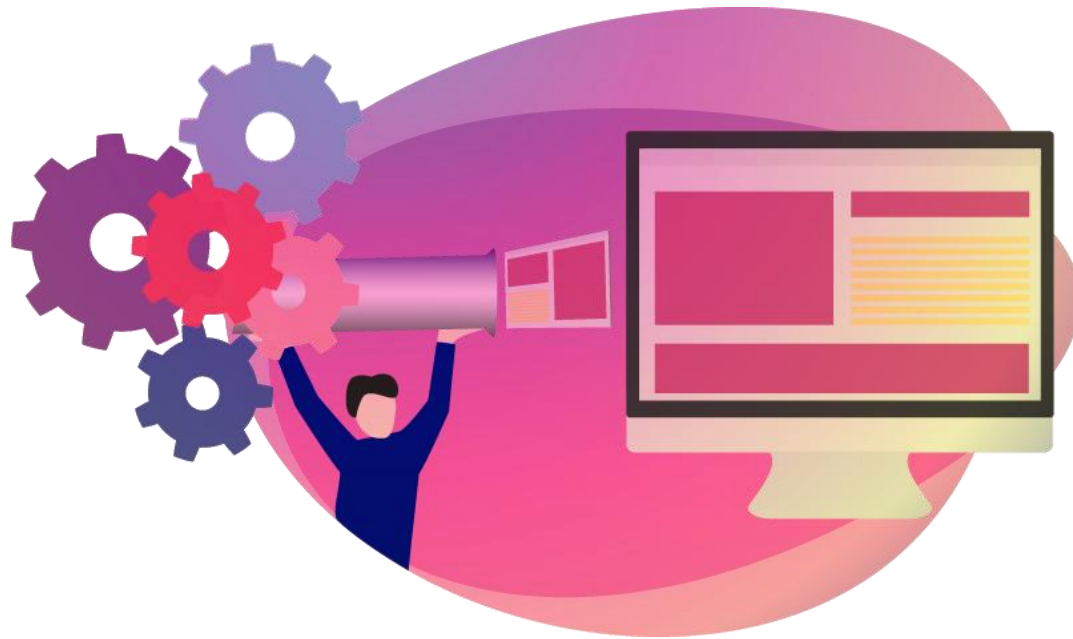


- Automatic sign-in is available for both on-premises and cloud-based applications.
- Users are not required to enter their passwords repeatedly.



Single Sign-On: Benefits

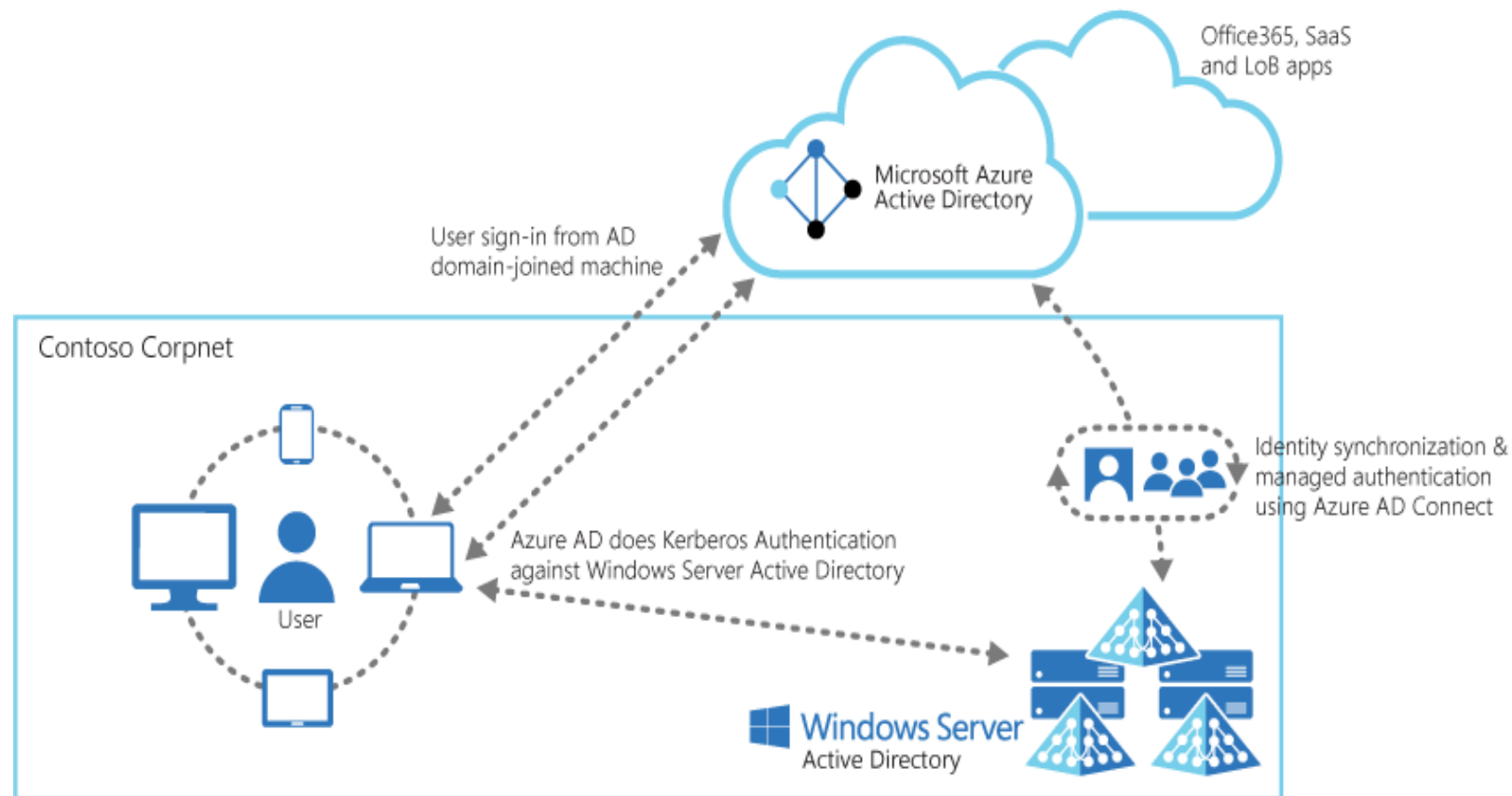
Easy deployment



- Needs no additional on-premises components
- Works with any method of cloud authentication
- Can be rolled out to some or all the users
- Registers non-Windows 10 devices with Microsoft Entra ID without the need for AD FS infrastructure

Single Sign-On: Features

The sign-in username can be the on-premises default username, or another attribute configured in the Microsoft Entra ID Connect method but cannot be used with AD FS.

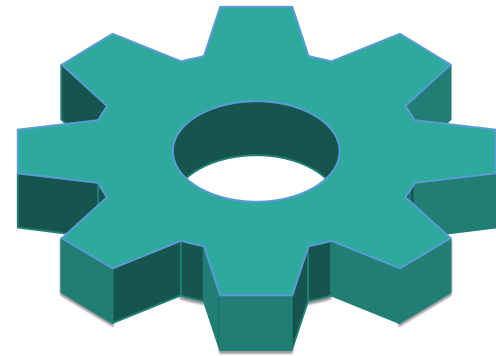


- If SSO fails, use a password at the sign-in page
- Sign in or out of other accounts
- Enabled via Microsoft Entra ID Connect
- Sample list of applications included with Microsoft Entra ID

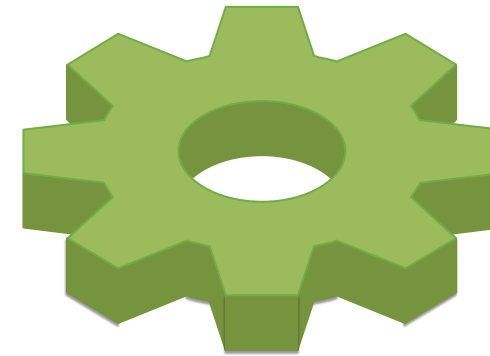
Here, Microsoft Entra ID is an alias for Microsoft Entra ID.

Microsoft Entra ID Seamless Single Sign-On: Considerations

They are:



It can be combined with either the Password Hash Synchronization or Pass-through Authentication sign-in methods.



Microsoft Entra Join provides SSO for devices registered with Microsoft Entra ID



Microsoft Entra ID Seamless Single Sign-On: Considerations

Applications using **domain_hint** or **login_hint** parameter capability of seamless SSO are:

Application name	Application URL to be used
Access panel	https://myapps.microsoft.com/contoso.com 
Outlook on Web	https://outlook.office365.com/contoso.com 
Office 365 portals	https://portal.office.com?domain_hint=contoso.com  , https://www.office.com?domain_hint=contoso.com 

Solution for Authentication

Authentication

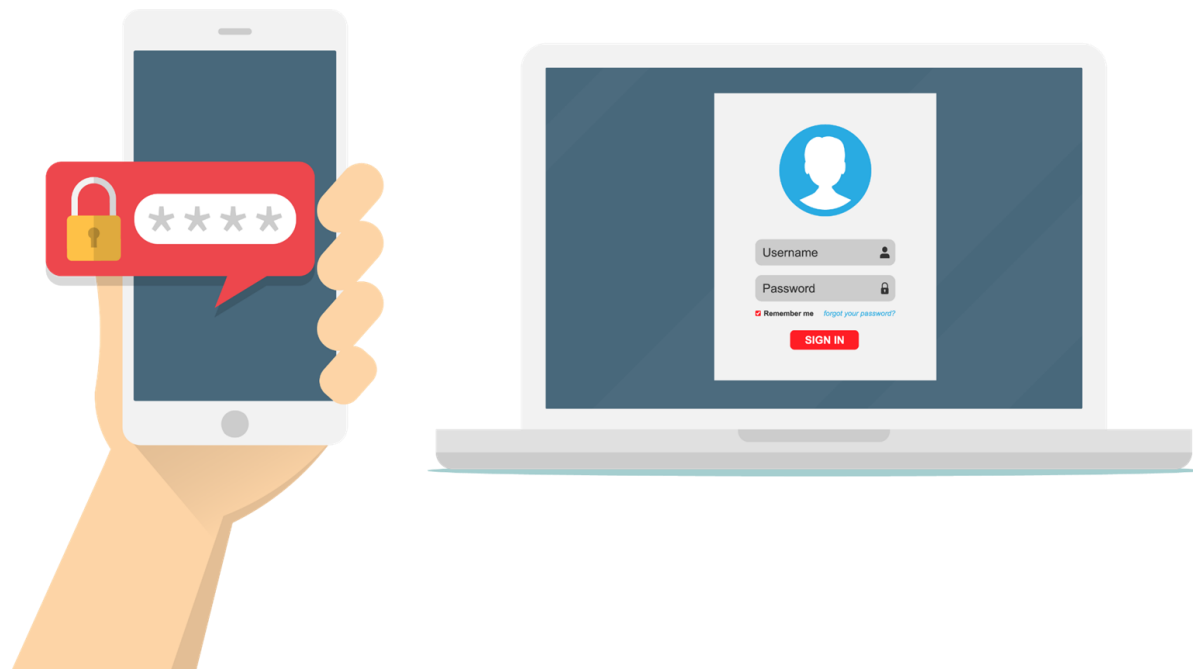
It is a process of confirming who the user claims to be.



The Microsoft identity platform implements the OpenID Connect protocol for handling authentication.

OAuth vs. OpenID Connect

OAuth is used for authorization and OpenID Connect (OIDC) is used for authentication.



OpenID Connect is built on top of OAuth 2.0 framework

It is possible to authenticate a user (using OpenID Connect) and get authorization to access a protected resource that the user owns (using OAuth 2.0) in one request.

Authentication Use Cases

The delegating authentication and authorization to Microsoft Entra ID enables scenarios such as:

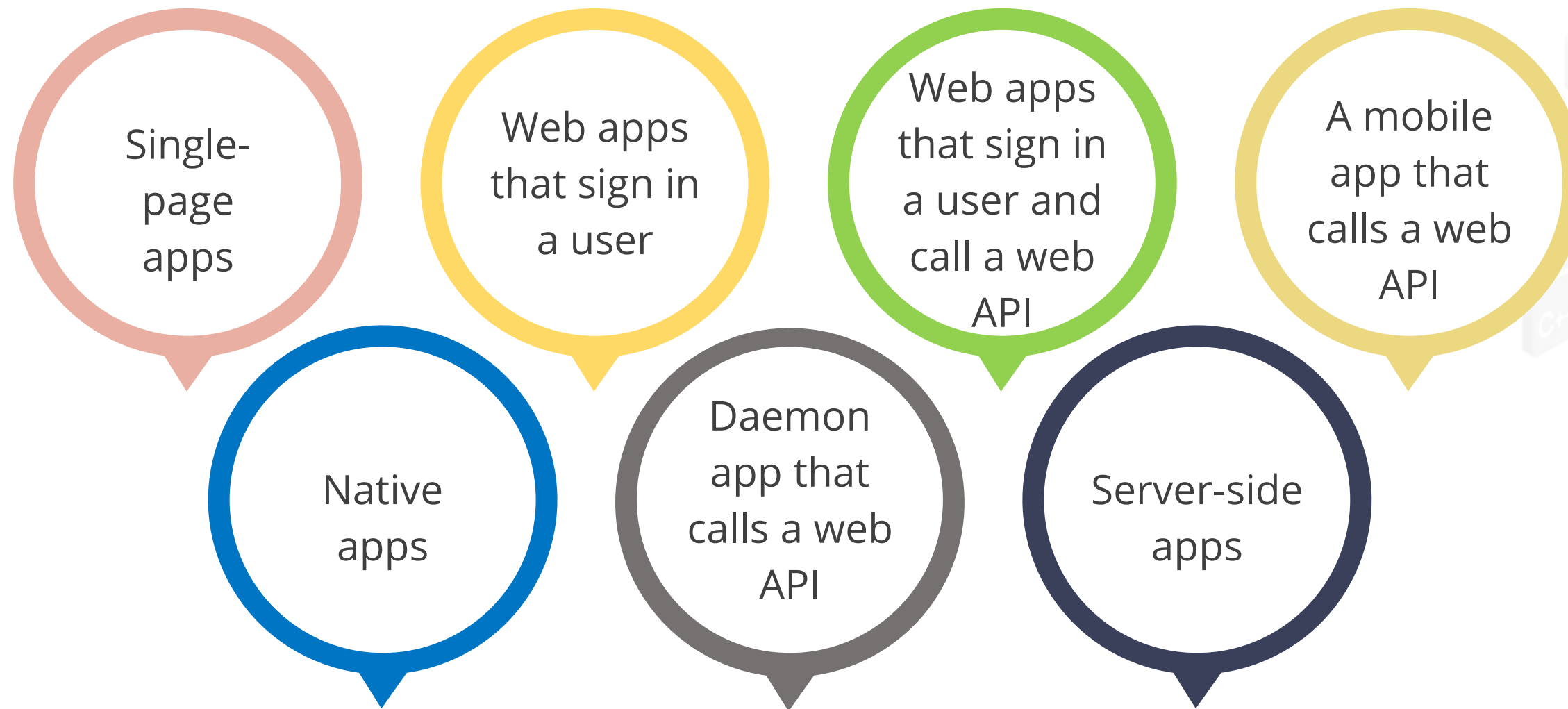
Conditional access policies that require a user to be in a specific location

SSO enables a user to sign in once and be automatically signed in to all the web apps

The use of multi-factor authentication, also called two-factor authentication or 2FA

Application Scenarios

The Microsoft identity platform supports authentication for these app architectures:



Solution for Conditional Access

Conditional Access

The policies are if-then statements, which means that if users wish to access a resource, they should first perform an action.



It allows a user to apply the appropriate access controls when needed to keep the organization secure.



Conditional Access

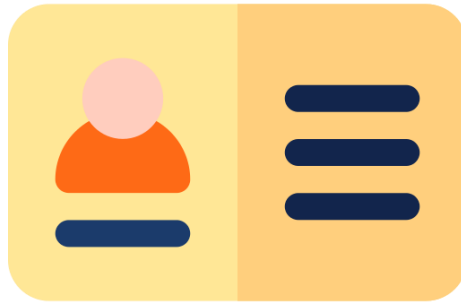
The following are the best practices for conditional access:



- Plan your costs
- Communicate with users and IT
- Use the zero-trust security model
- Establish the steering committee

Multi-Factor Authentication (MFA)

It is a process where a user is prompted during the sign-in process for an additional form of identification.



Username or
password



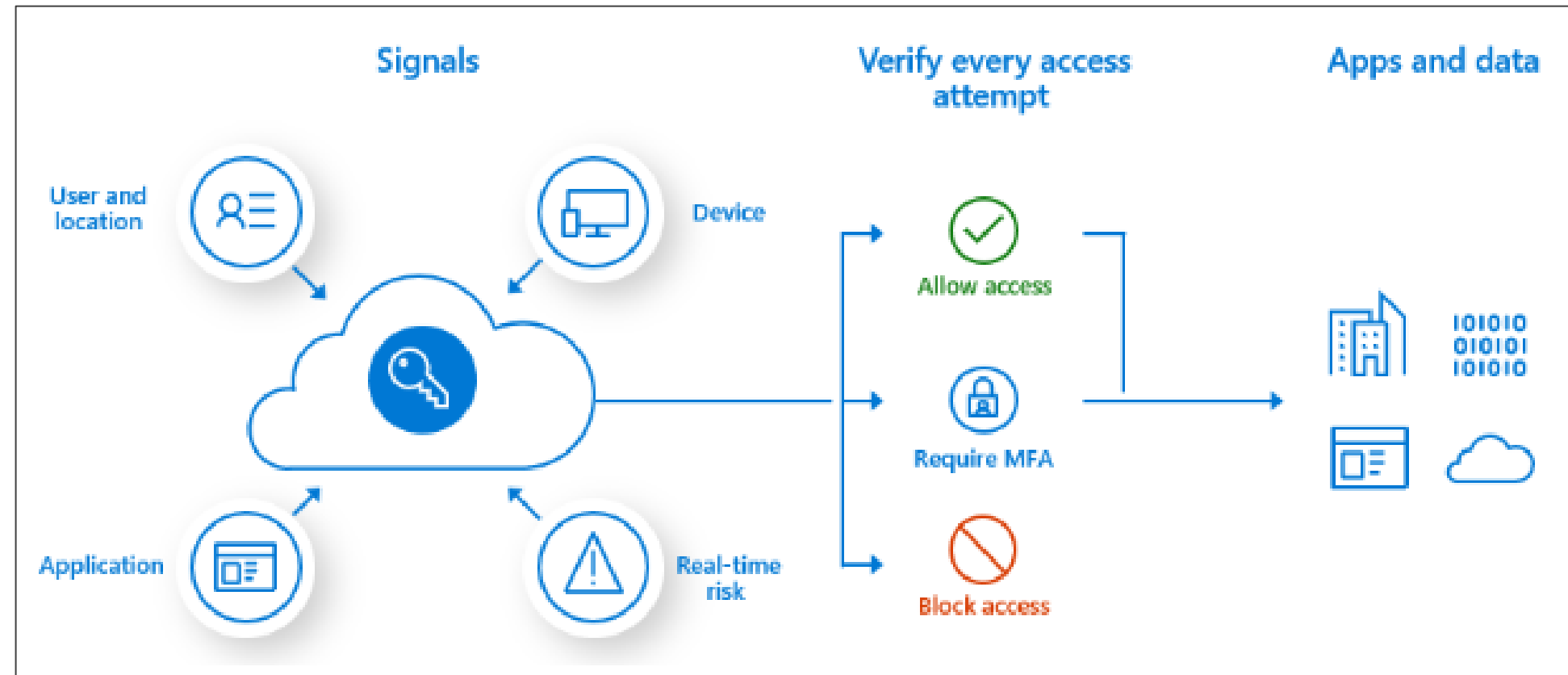
Phone or
hardware key



Biometrics like
fingerprint or face
scan

Multi-Factor Authentication (MFA)

It provides two-step authentication and verification.



Source: <https://learn.microsoft.com/en-us/entra/identity/authentication/concept-mfa-howitworks>

MFA Authentication Methods

These are the authentication methods of MFA:

Authentication method	Security	Usability	Availability
Windows Hello for Business	High	High	High
Microsoft Authenticator	High	High	High
Authenticator Lite	High	High	High
FIDO2 security key	High	High	High
Certificate-based authentication	High	High	High
OATH hardware tokens (preview)	Medium	Medium	High
OATH software tokens	Medium	Medium	High
Temporary Access Pass (TAP)	Medium	High	High
SMS	Medium	High	Medium
Voice	Medium	Medium	Medium
Password	Low	High	High



Plan for MFA Deployment

Below are the deployment considerations:

- All users, a specific user, a group member, or a role allocated
- Device platform
- State of the device
- Client applications
- Hybrid Microsoft Entra ID joined device



Conditional Access and Azure MFA

Microsoft Entra MFA allows a user to impose restrictions on app access depending on the conditions listed below:



- MFA can be set for users and groups to prompt additional verification during sign in.
- Conditional access policies can be used to establish MFA-required events or applications.

Configure MFA Settings

The MFA settings are given in the table below:

Feature	Description
Account lockout	Temporarily locks accounts if there are too many denied authentication attempts in a row
Block/unblock users	Blocks specific users from being able to receive MFA requests
Report suspicious activity	Configures settings that allow users to report fraudulent verification requests
Notifications	Enables notifications of events from the MFA server
OATH tokens	Used in cloud-based Azure MFA environments to manage OATH tokens for users
Phone call settings	Configures settings related to phone calls and greetings for cloud and on-premises environments
Providers	Shows any existing authentication providers that may have associated with an Azure account

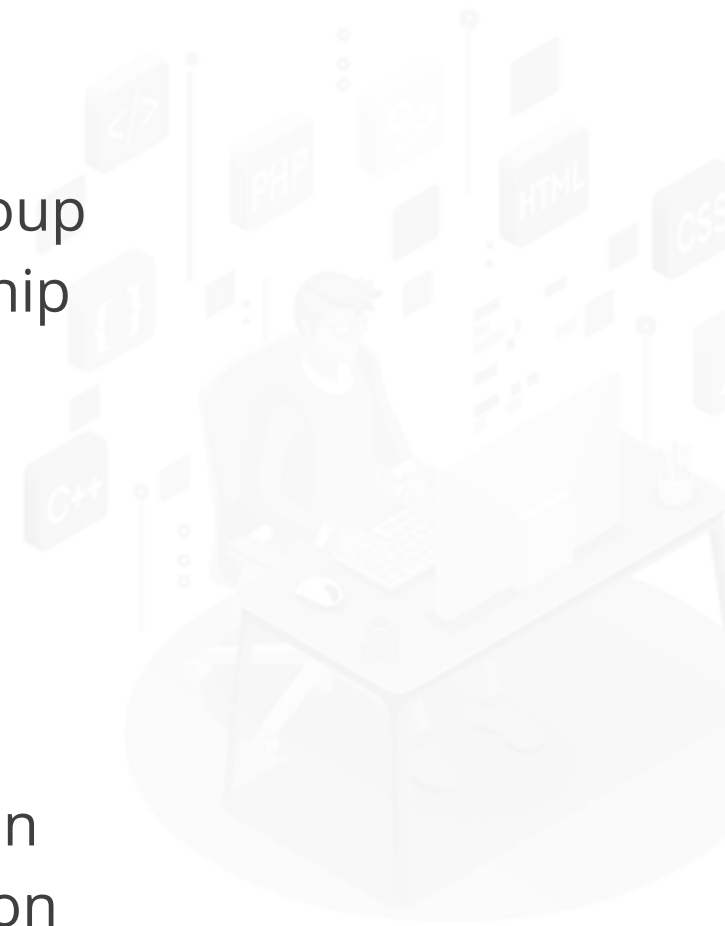
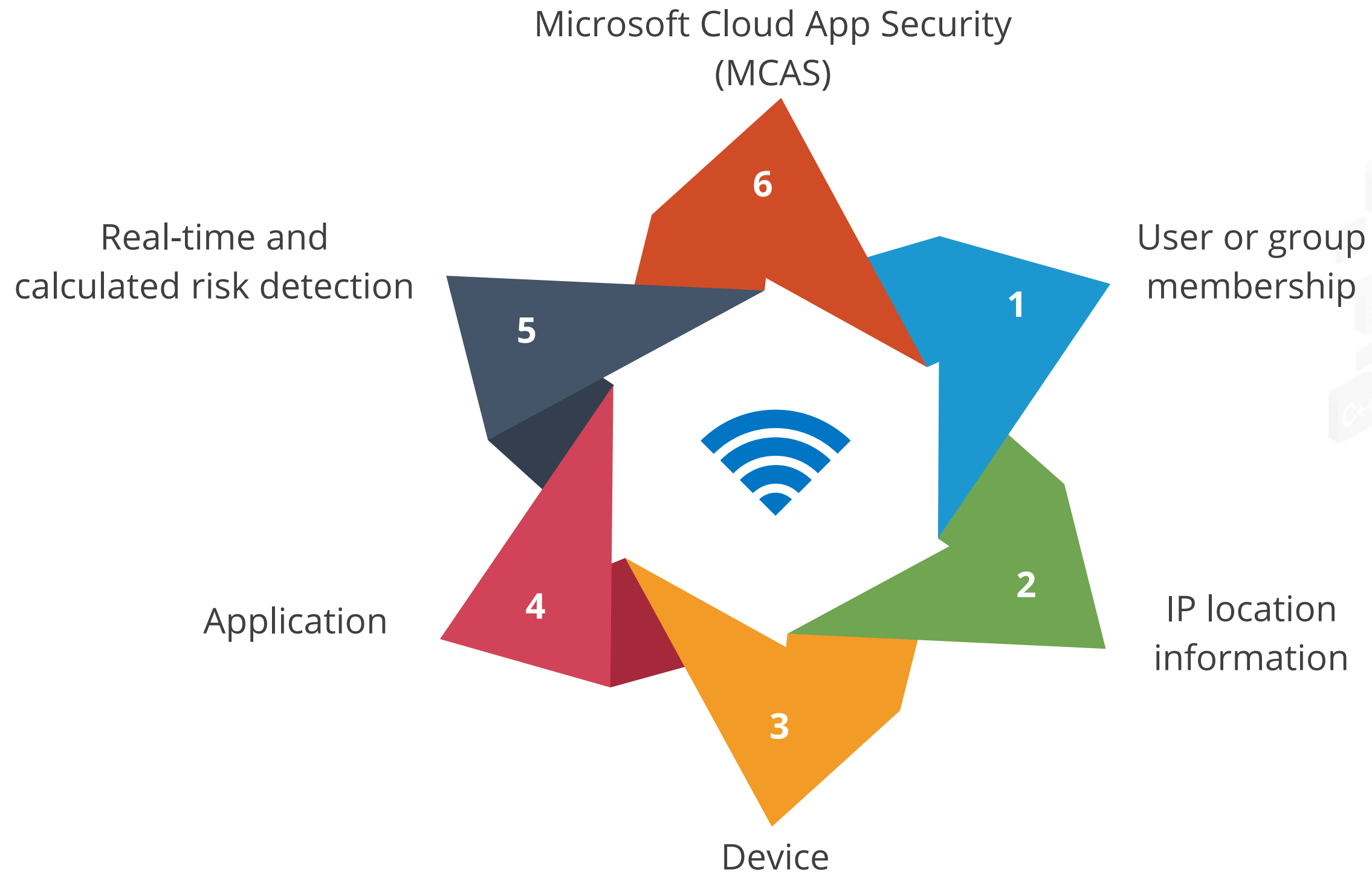
Conditional Access: Commonly Applied Policies



- Requires MFA for users with administrative roles
- Requires MFA for Azure management tasks
- Blocks or grants access for users attempting to use legacy authentication protocols
- Requires trusted locations for security information registration
- Blocks or grants access from specific locations
- Requires organization-managed devices for specific applications
- Blocks risky sign-in behaviors



Conditional Access: Common Signals



Conditional Access: Common Decisions

The following are the common decisions that conditional access should consider while making a policy decision:



Conditional Access: Common Decisions

Grant access

- Enforce additional security by requiring users to verify their identity with multiple factors
- Specify the strength of authentication methods to access resources.
- Ensure devices meet compliance policies before granting
- Limit access to specific client apps that meet security and compliance standards

Block access

- A robust control mechanism used to prevent access to resources.
- Requires careful consideration and thorough testing to avoid unintended consequences.
- Use Conditional Access tools such as Report-Only Mode and What If Tool for simulation and validation before implementation.

Access Review

It enables organizations to efficiently manage group memberships, access to enterprise applications, and role assignments, and it requires P2 subscription.

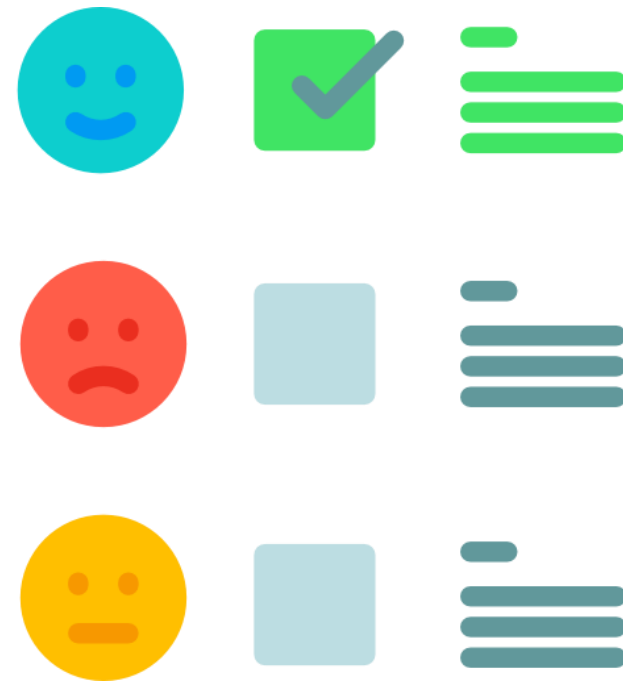


The user's access can be reviewed on a regular basis to make sure only the right people have continued access.



Importance of Access Review

An access review is a Microsoft Entra ID Governance capability.



Planning the access review deployment is essential to make sure that users achieve the desired governance strategy.



Benefits of Access Review

They are as follows:



Benefits of Access Review

They are as follows:

Control collaboration

It allows users to manage access to all the resources. When users share and collaborate, one can be assured that the information is among authorized users only.

Manage risk

It provides a way to review access to data and applications that lowers the risk of data leakage and data spill.

Benefits of Access Review

They are as follows:

Address compliance and governance

It allows users to govern and recertify the access lifecycle to groups, apps, and sites. Users can also control and track reviews for compliance or risk-sensitive applications.

Reduce cost

It costs less than building or upgrading on-premises tools, as access reviews optimize resource usage, manage risks, and ensure compliance.

Solution That Includes Managed Identities

Azure Managed Identity

It provides an automatically managed identity for applications to connect with resources that support Microsoft Entra authentication.



It can be used to authenticate to services that support Microsoft Entra authentication.

Azure Managed Identity

The benefits are:

- No need to manage credentials
- Any resource can be authenticated
- No additional cost
- No need for rotating credentials or certificates



Azure Managed Identity Terminologies

Managed identities involve the use of these terms:

Client ID

A unique ID linked to the Microsoft Entra ID application and service principal

Object ID

The service principal object of the managed identity

Azure Instance Metadata Service

A REST API that's enabled when Azure Resource Manager provisions a VM

Service Principal



- It is an identity created for use with applications, services, and automated processes.
- It is used to grant permissions and access to resources.
- It is often used in automation, scripts, and service-to-service communication.

When setting up an application or service to access Azure resources, create a service principal to provide secure authentication.

Security Principal

It encompasses users and service principals and refers to any entity authorized or authenticated to access resources.



- Users with Entra ID accounts and service principals both belong to the category of security principals.
- Security principals, whether users or service principals, can have roles and permissions assigned, defining their actions within Azure resources.

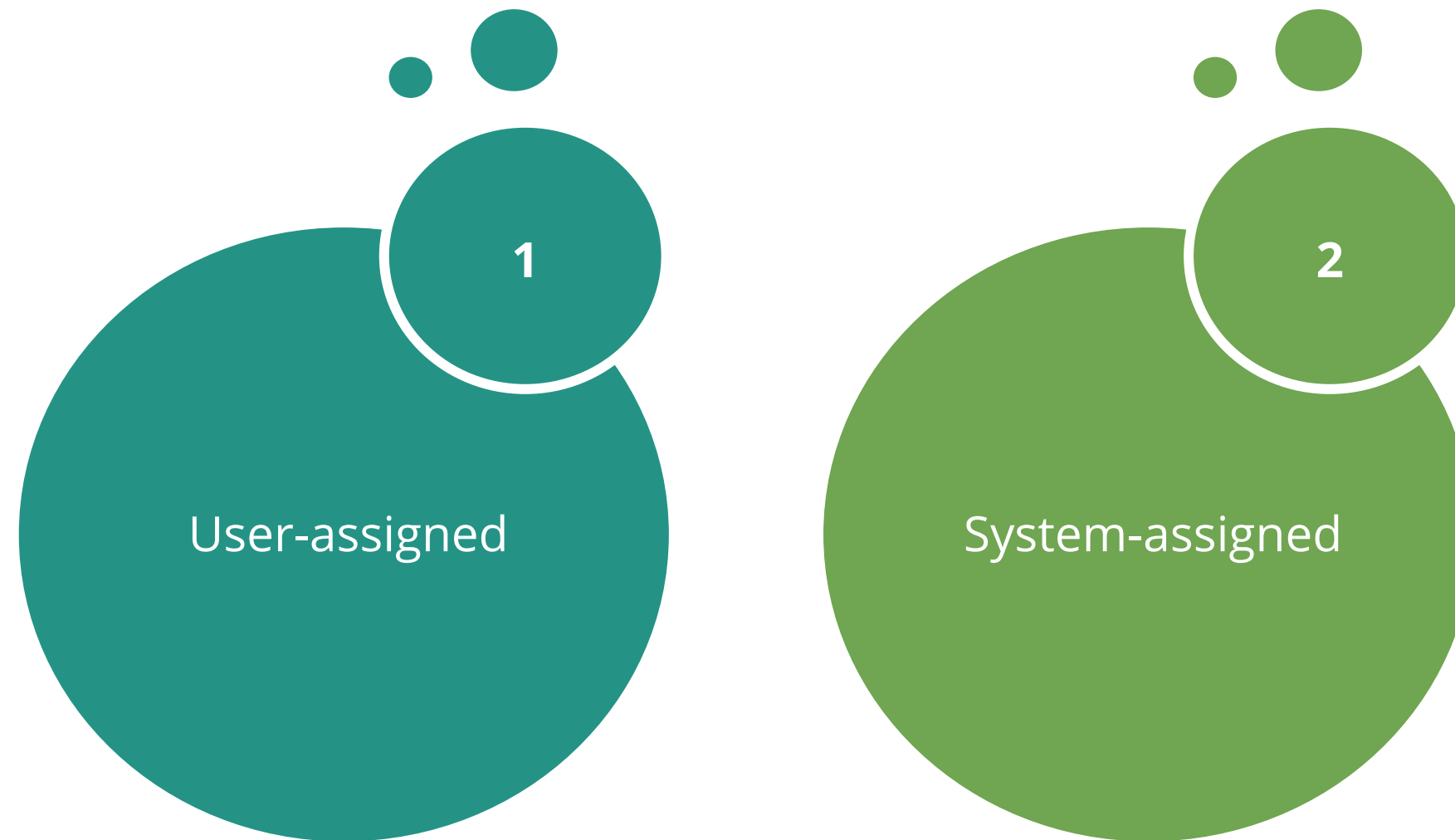
Difference Between Service Principal and User

Service principal	User (a type of security principal)
Represents applications, services, or automation	Represents individual users with Entra ID accounts
Typically used for non-human entities	Used for human access and interaction with Azure resources
Often used for headless authentication and automated tasks	Requires manual authentication by users



Types of Managed Identities

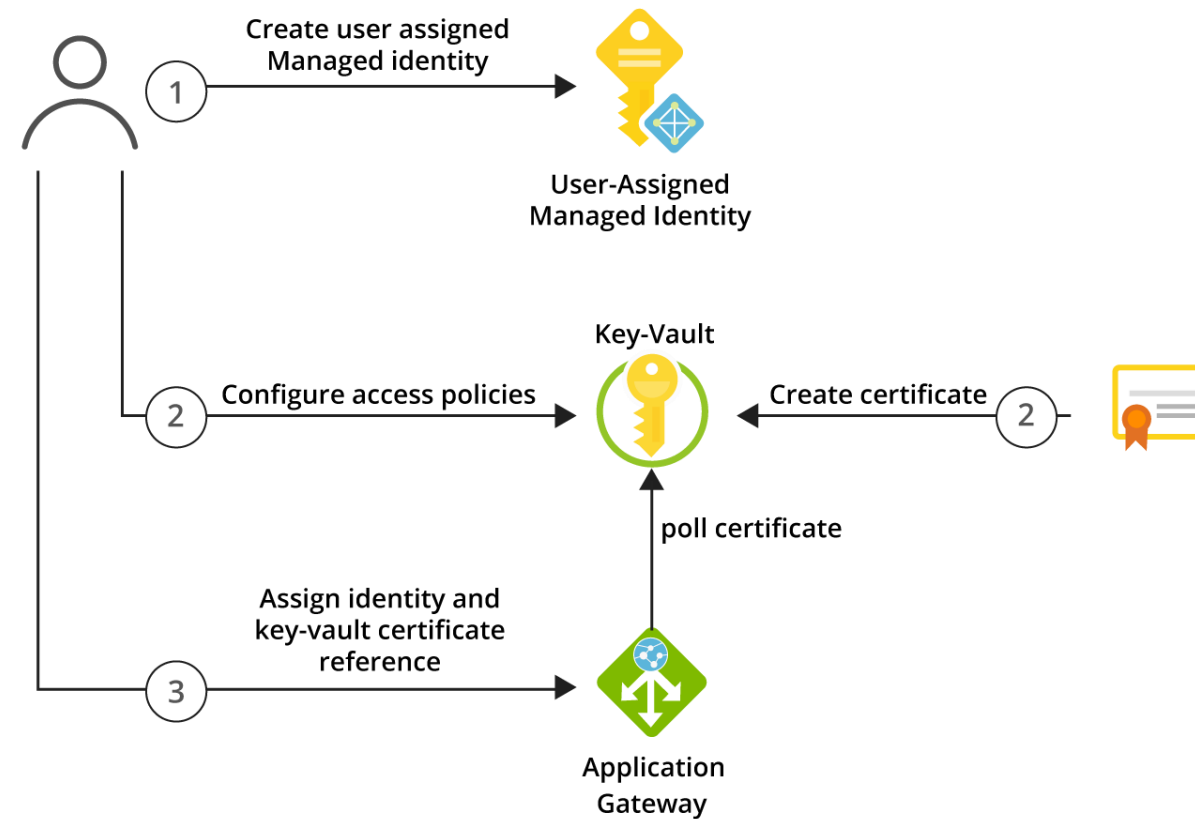
There are two types of managed identities:



User-Assigned Managed Identity

Azure creates a service principal just as it does for a system-assigned identity.

It is created as a standalone Azure resource.



System-Assigned Managed Identity

When a user enables the identity, Azure creates a service principal through Azure Resource Manager.



System-assigned managed identity shares the same lifecycle as the VM.



Selecting the Appropriate Managed Identity

The following table shows the differences between the two types of managed identities:

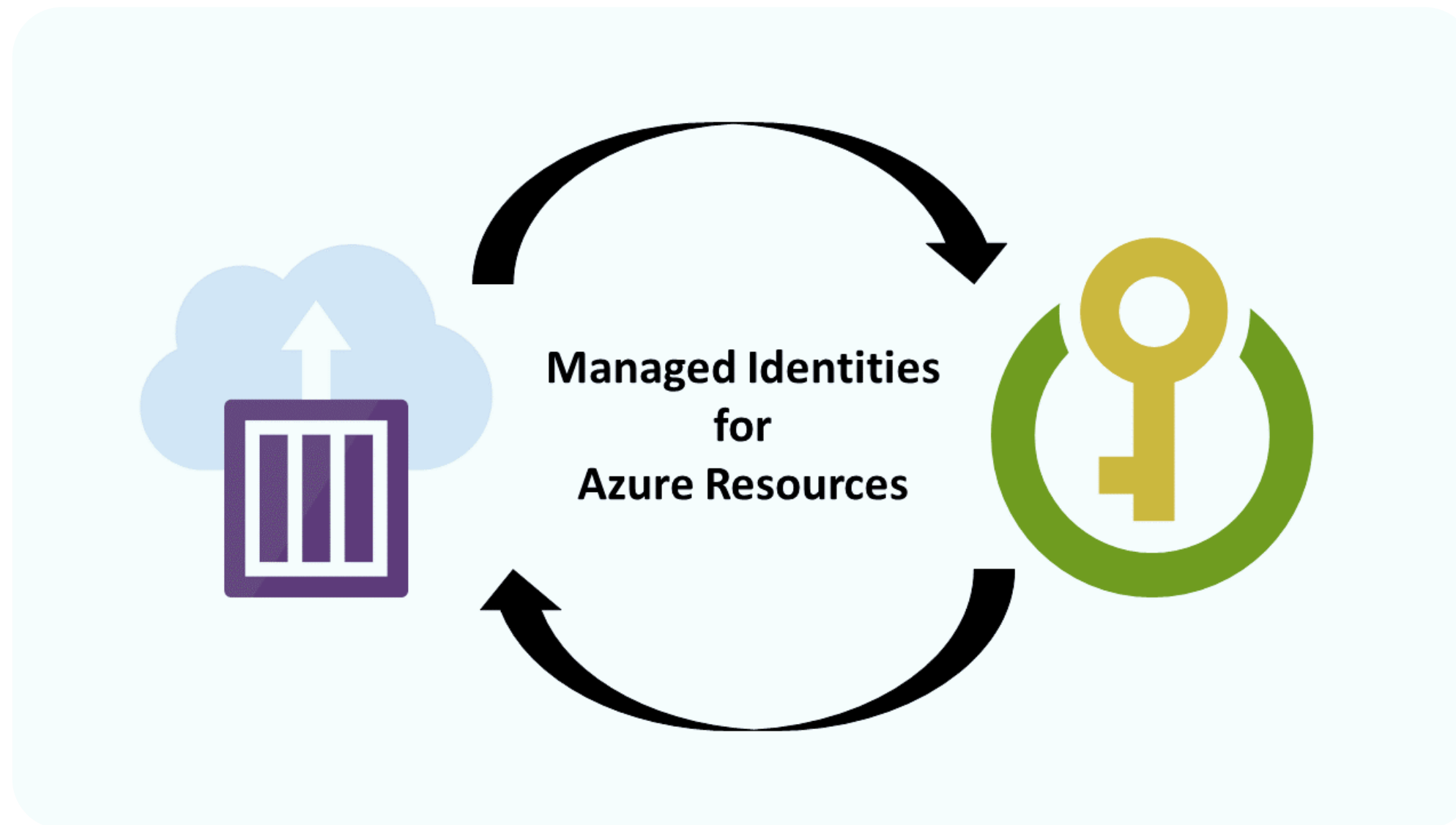
Property	System-assigned managed identity	User-assigned managed identity
Creation	Created as part of an Azure resource	Created as a stand-alone Azure resource
Life cycle	Shared life cycle with the Azure resource	- Independent life cycle - Must be explicitly deleted
Sharing across Azure resources	- Cannot be shared - Can only be associated with a single Azure resource	- Can be shared - Can be associated with more than one Azure resource

Selecting the Appropriate Managed Identity

Property	System-assigned managed identity	User-assigned managed identity
Common use cases	• Workloads that are contained within a single Azure resource • Workloads for which independent identities are needed • For example, an application that runs on a single virtual machine	• Workloads running on multiple resources, which can share a single identity • Workloads requiring pre-authorization to a secure resource in the provisioning flow • Workloads with frequent resource recycling, maintaining consistent permissions

Managed Identity for Azure Resources

It manages the credentials to authenticate cloud services when building cloud applications.

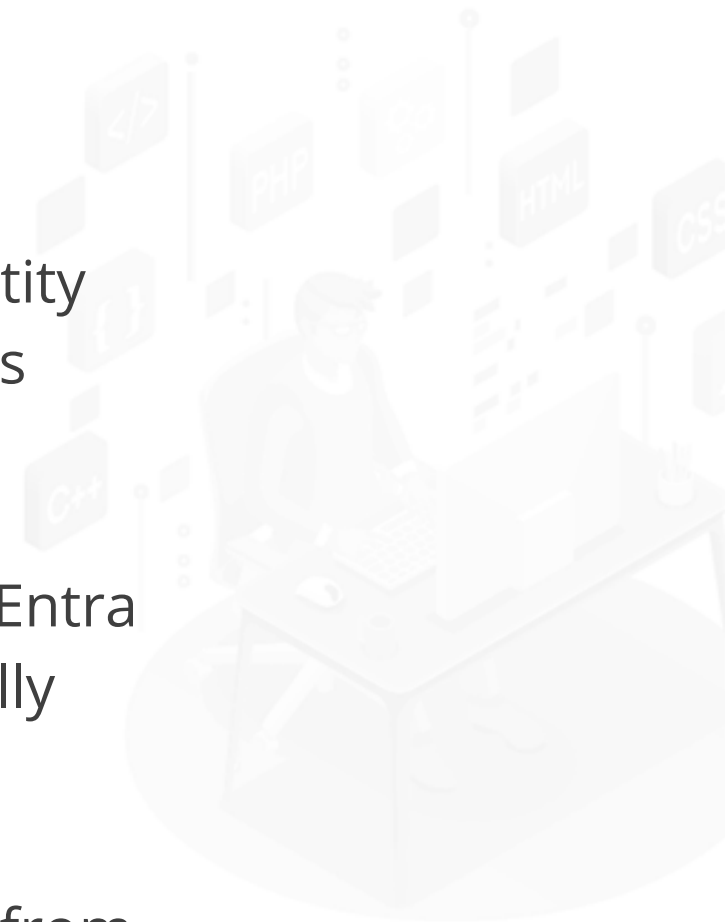


Managed Identity for Azure Resources

These are the advantages of managed identity when used for Azure resources:

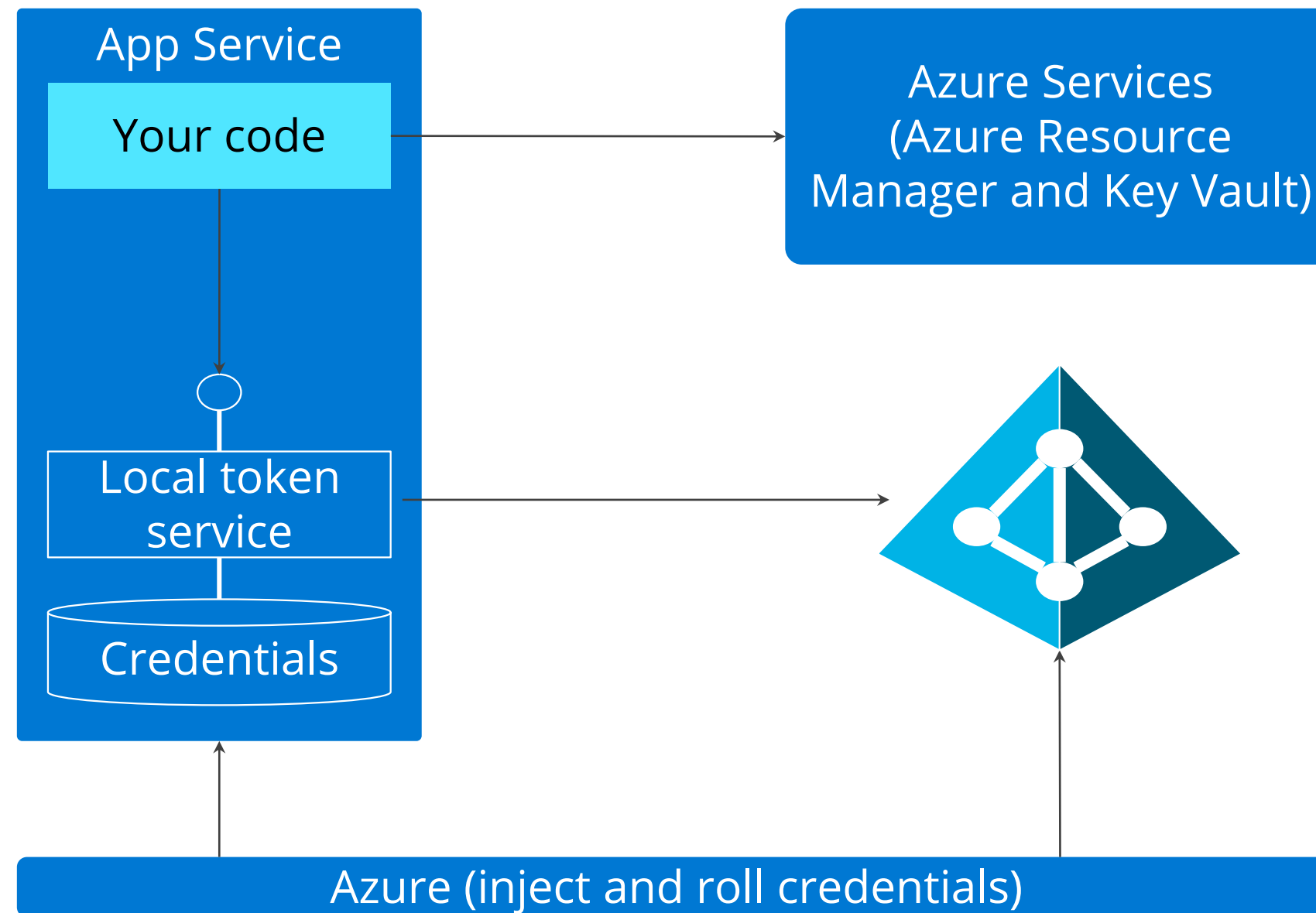


- Keeps credentials out of code
- Uses a local Managed Service Identity (MSI) endpoint to get access tokens from Microsoft Entra ID
- Manages the identity in Microsoft Entra ID for Azure resources automatically
- Offers direct authentication with services or retrieval of credentials from the Azure Key Vault



Workflow of Managed Identity

The following diagram shows the internal workflow of managed identity:



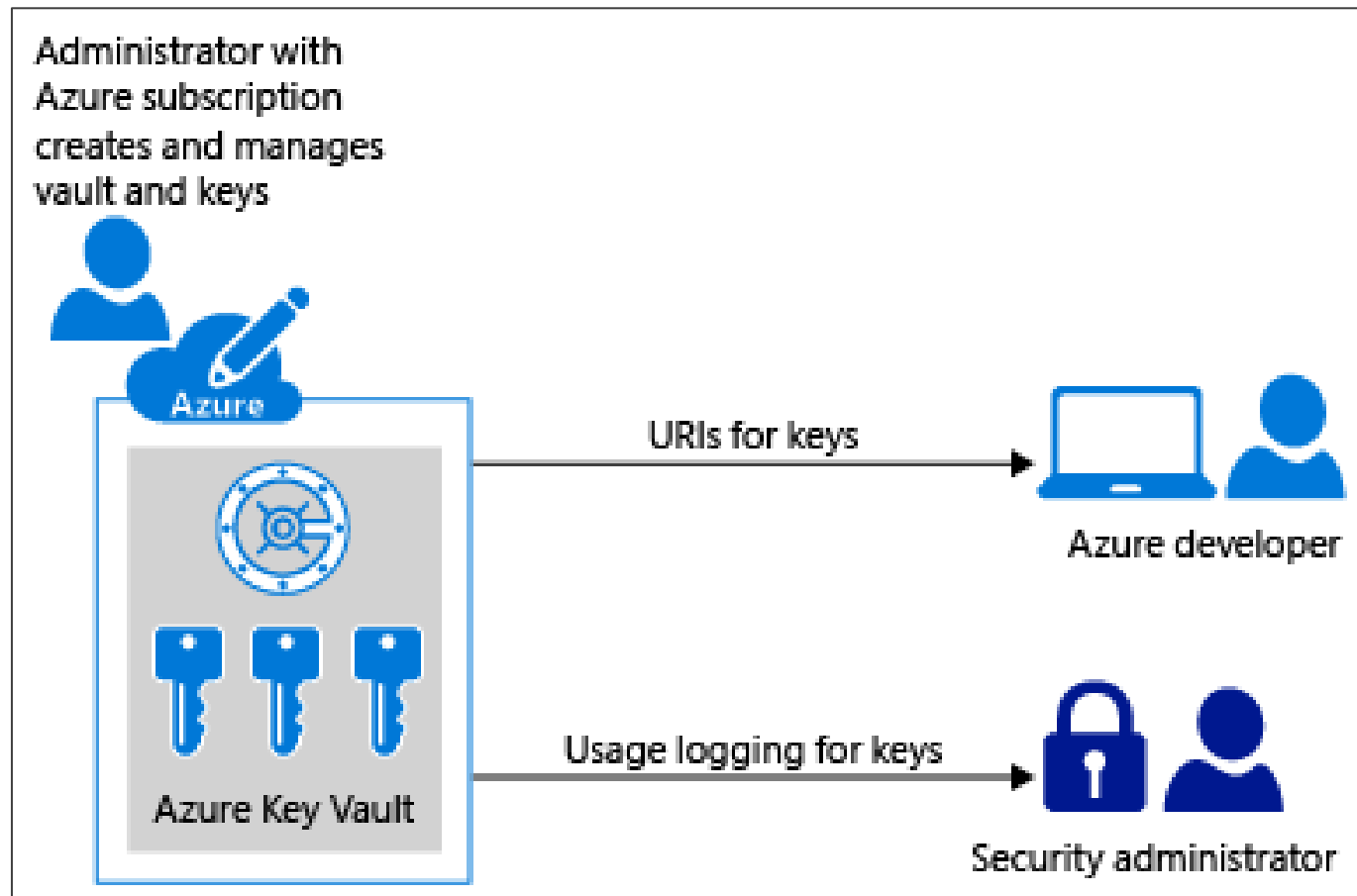
Solution That Includes Key Vault

Azure Key Vault

It is a centralized cloud service for storing application secrets such as encryption keys, certificates, and server-side tokens.



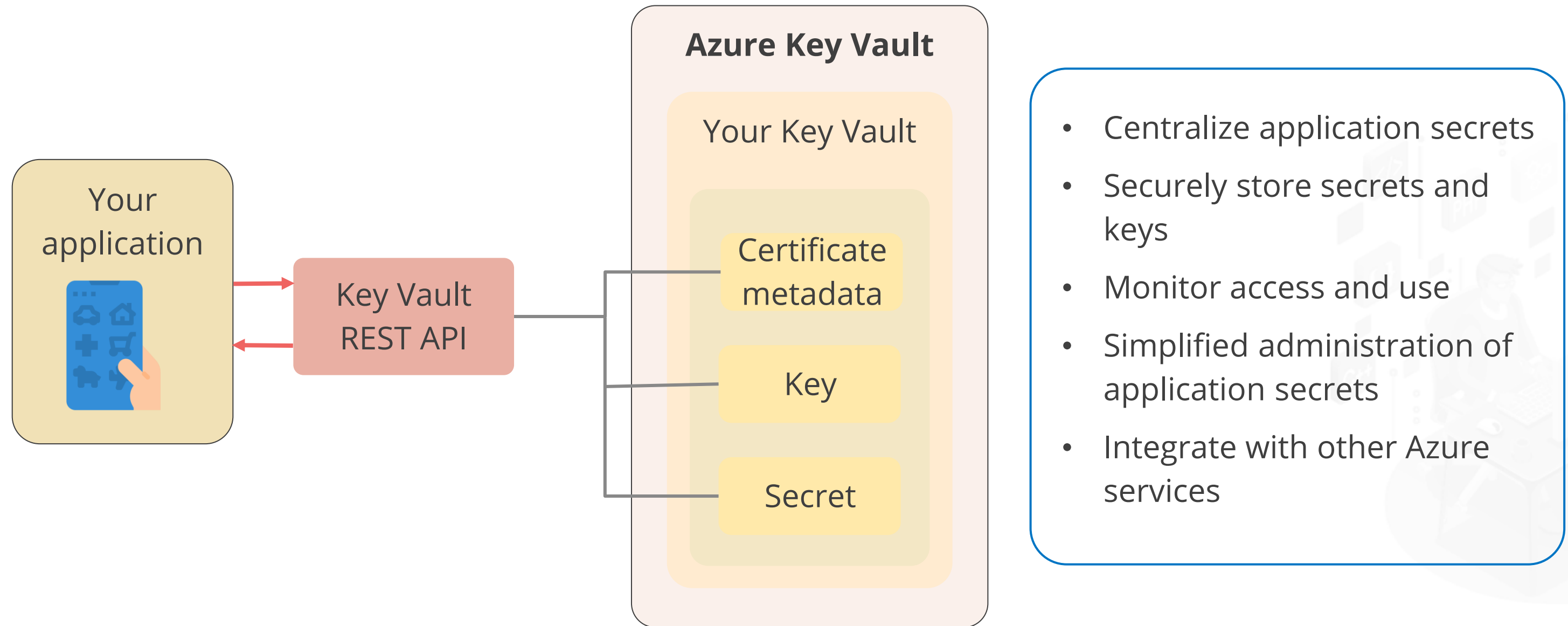
Azure Key Vault



Azure Key Vault offerings:

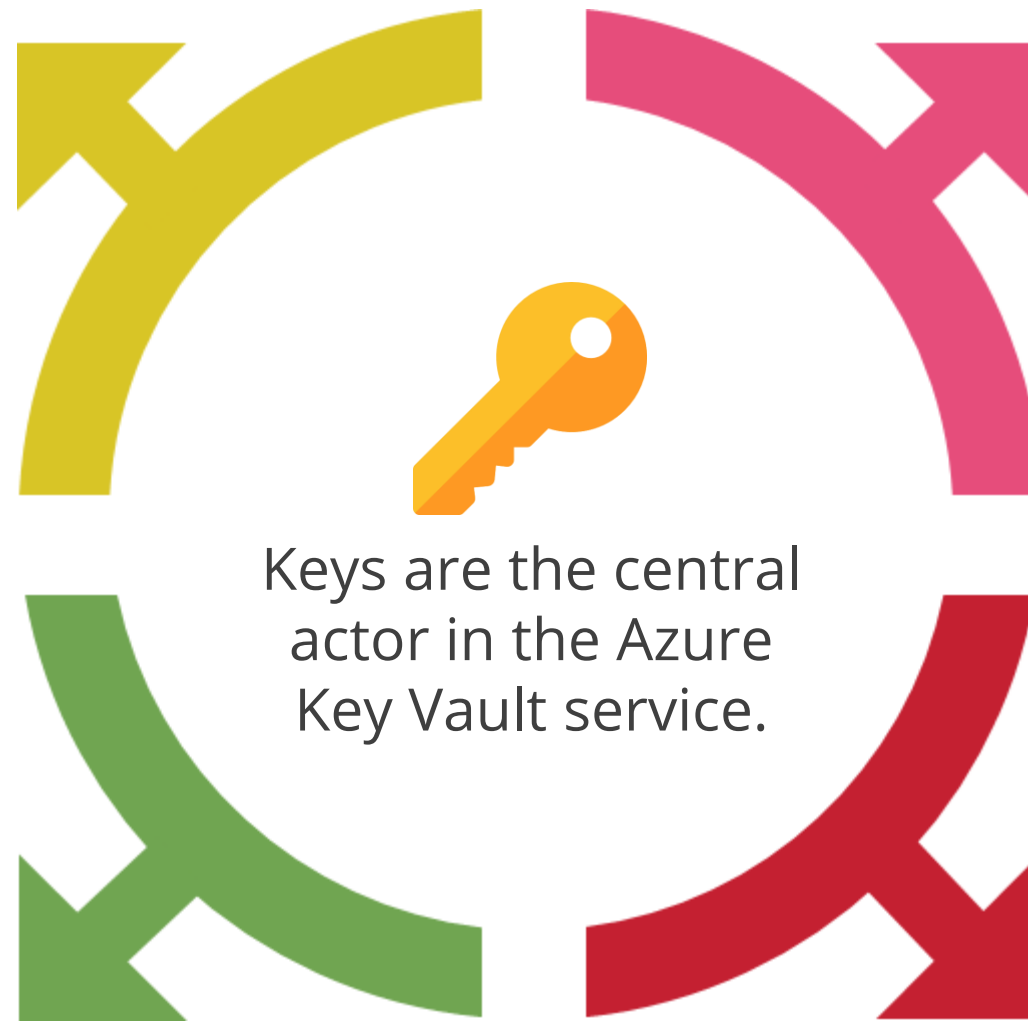
- Secrets management
- Key management
- Certificate management
- Secret storage

Azure Key Vault Benefits



Keys

They can be single instanced (only one key exists).



They are cryptographic assets that serve a specific purpose.

They can be versioned (mix of primary and secondary keys).

They are never directly accessible to applications.



Keys

Two types of keys in Key Vault

Hardware-protected

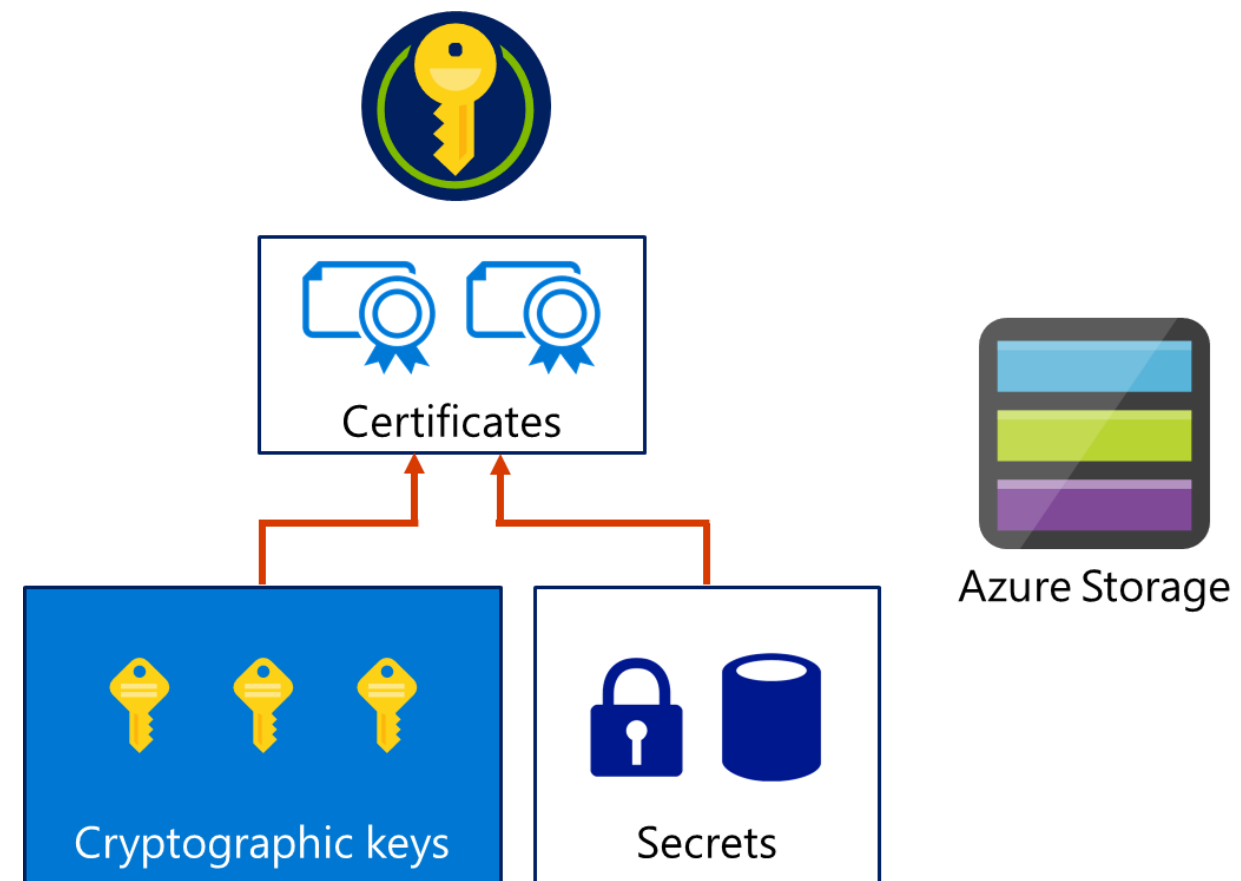
Supports using hardware security module (HSMs) that provide a hardened, tamper-resistant environment for cryptographic processing and key generation

Software-protected

Supports using software-based Rivest–Shamir–Adleman (RSA) and elliptic curve cryptography (ECC) algorithms

Secrets

Key Vault stores and manages secrets as sequences of octets (8-bit bytes) with a maximum size of 25k bytes each.



Types of Keys

Cryptographic keys

The Key Vault supports multiple key types and algorithms and using hardware security modules (HSMs) for high-value keys.

Certificates

The Key Vault supports certificates, which are built on top of keys and secrets and add an automated renewal feature.

Secrets

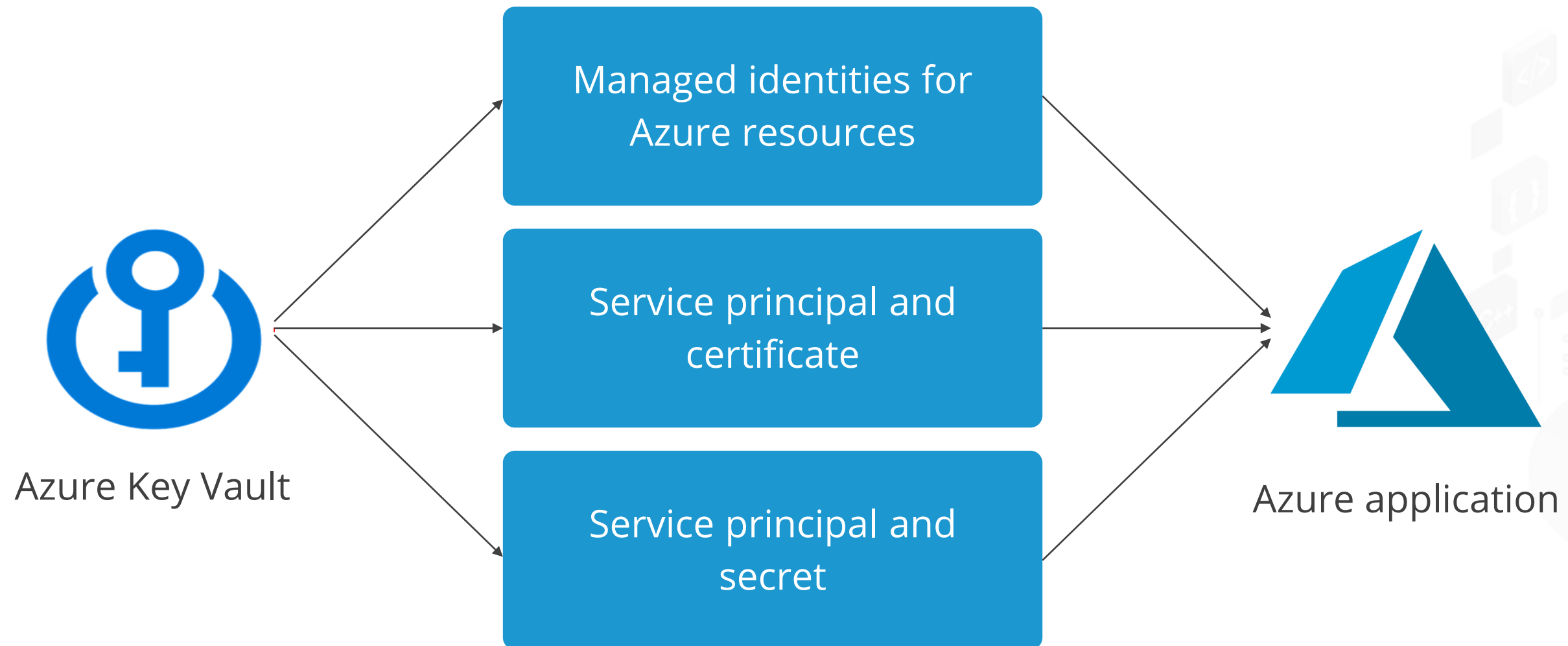
The Key Vault provides secure storage of secrets, such as passwords and database connection strings.

Azure storage

The Key Vault can manage the keys of an Azure Storage account. Internally, Key Vault can list (sync) keys with an Azure Storage Account and regenerate (rotate) the keys periodically.

Key Vault Authentication and Authorization

Illustration of authentication and authorization between the Azure Key Vault and Azure application:



Key Vault Authentication and Authorization

The procedure for accessing the Key Vault:

Authenticate

Microsoft Entra ID to authenticate users and applications

Read and write data in the Azure Key Vault

Uses a separate Key Vault access policy



Restrict network access

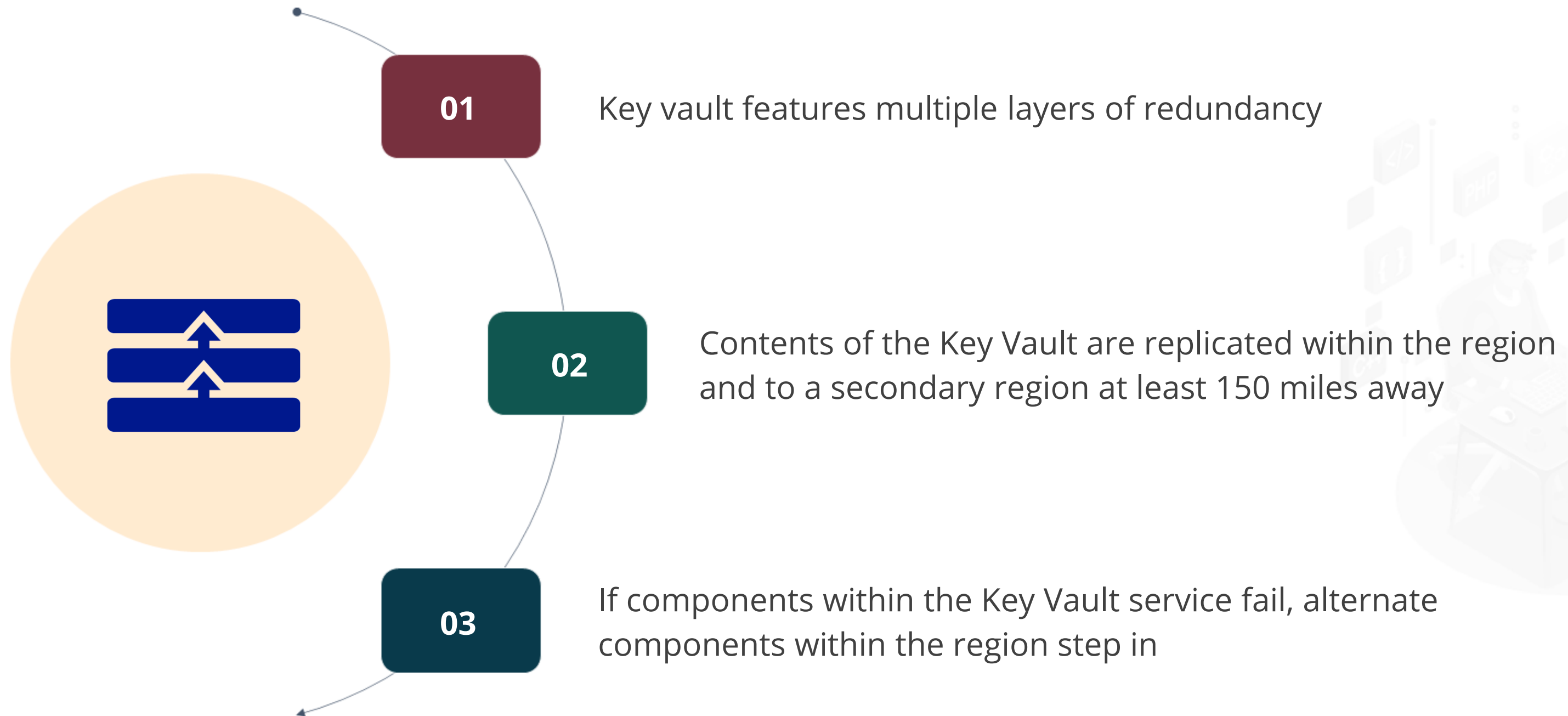
Services in the network can access the vault

Authorize

Role-based access control (RBAC)

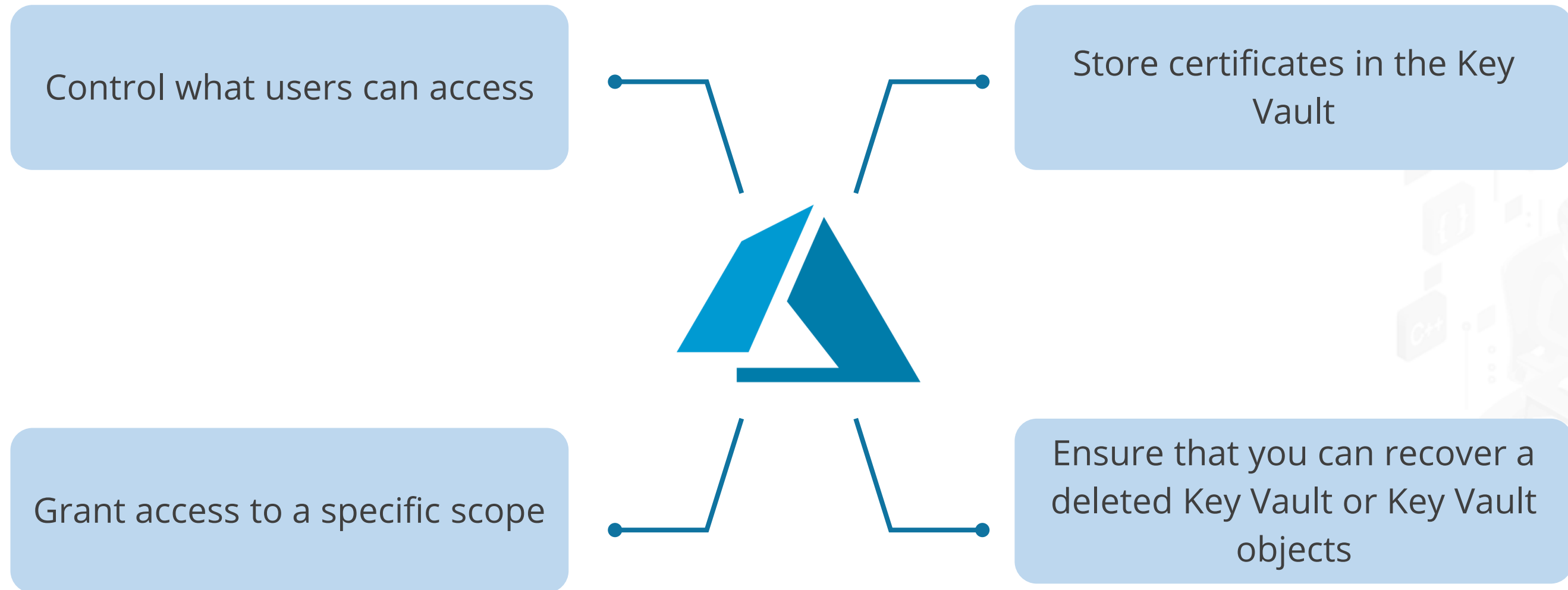
Key Vault Availability and Redundancy

Availability and redundancy are distinct features in the Azure Key Vault:



Key Vault Best Practices

The best practices of the Key Vault:



Creating an Azure Key Vault



Duration: 10 Min.

Problem Statement:

As an Azure Architect, you've been asked to provide your organization with an Azure security solution for storing application secrets such as tokens, passwords, certificates, API keys, and other secrets such as certificates, keys, and secret management.

Outcome:

You will be able to create and configure an Azure Key Vault to securely store application secrets such as tokens, passwords, certificates, API keys, and manage these secrets effectively.

Note: Refer to the demo document for detailed steps:
[01_Creating_an_Azure_Key_Vault](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Create an Azure Key Vault

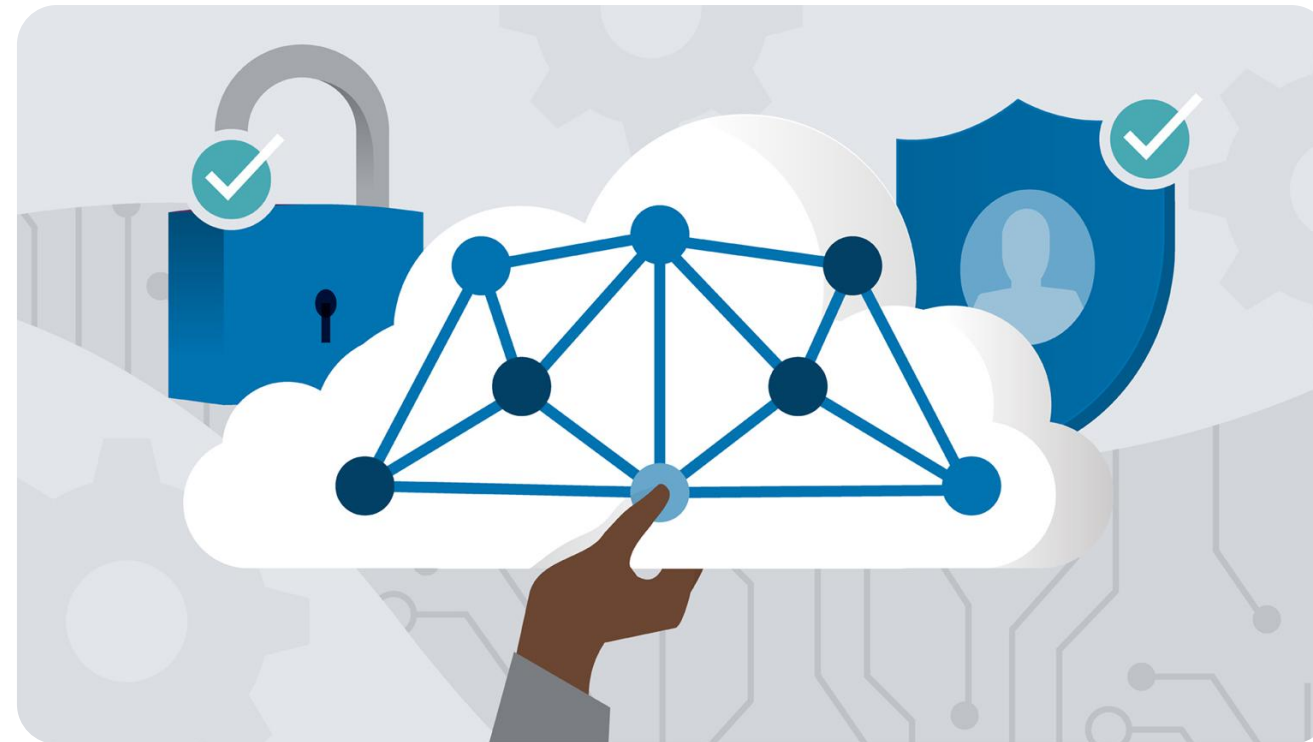


TECHNOLOGY

Solution for Hybrid Identity

Hybrid Identity

It seamlessly integrate on-premises and cloud capabilities, establishing a unified user identity for authentication and authorization across all resources, regardless of location.



Hybrid Identity

The following authentication methods can be used to implement hybrid identity with Microsoft Entra ID:

Password hash
synchronization
(PHS)

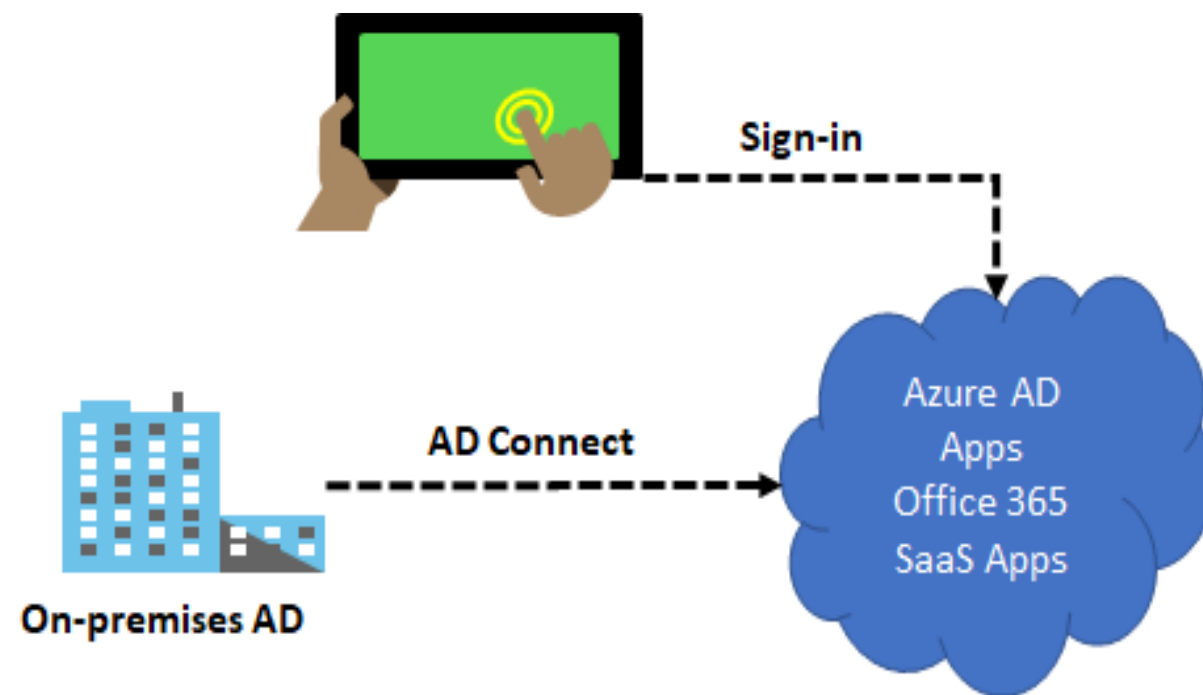
Pass-through
authentication
(PTA)

Federated
authentication



Microsoft Entra Connect

Integrating on-premises directories with Microsoft Entra ID provides a common identity for accessing both cloud and on-premises resources.



- Users can use a single identity to access on-premises applications and cloud services.
- Single tool to provide an easy deployment experience for synchronization and sign-in.

Microsoft Entra ID Connect replaces older versions of identity integration tools such as DirSync and Microsoft Entra ID Sync.

Common Scenarios

Below are the common scenarios with recommended hybrid identity options:

User needs to:	PHS and SSO	PTA and SSO	Federation
Sync new user, contact, and group accounts created in their on-premises Active Directory to the cloud automatically	X	X	X
Set up tenant for Office 365 hybrid scenarios	X	X	X
Enable users to sign in and access cloud services using their on-premises password	X	X	X

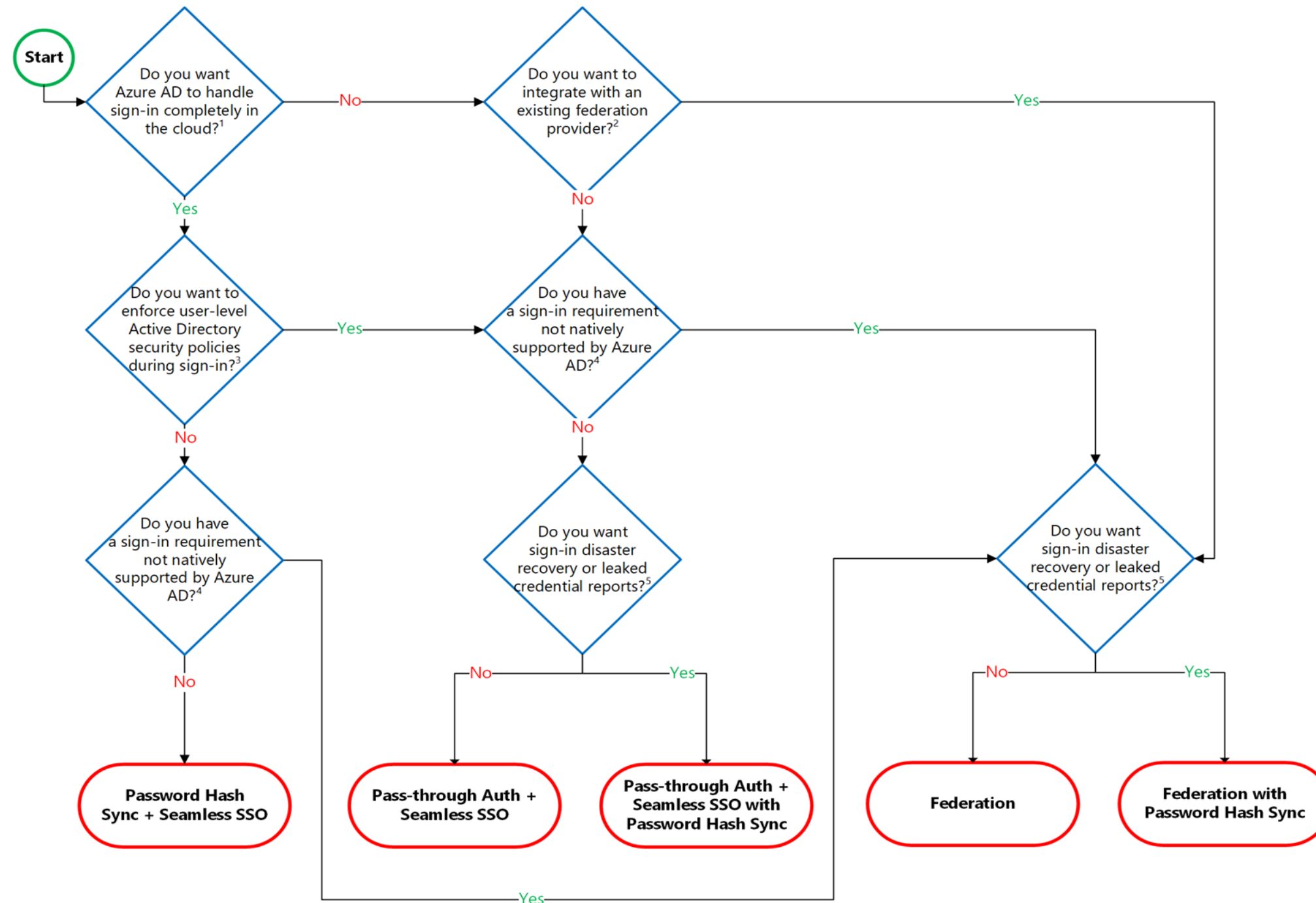
Common Scenarios

Below are the common scenarios with recommended hybrid identity options:

User needs to:	PHS and SSO	PTA and SSO	Federation
Implement single sign-on using corporate credentials	X	X	X
Ensure no password hashes are stored in the cloud		X	X
Enable cloud-based multi-factor authentication solutions	X	X	X
Enable on-premises multi-factor authentication solutions			X
Support smart card authentication for users			X

Hybrid Identity Decision Tree

The workflow is shown as below:

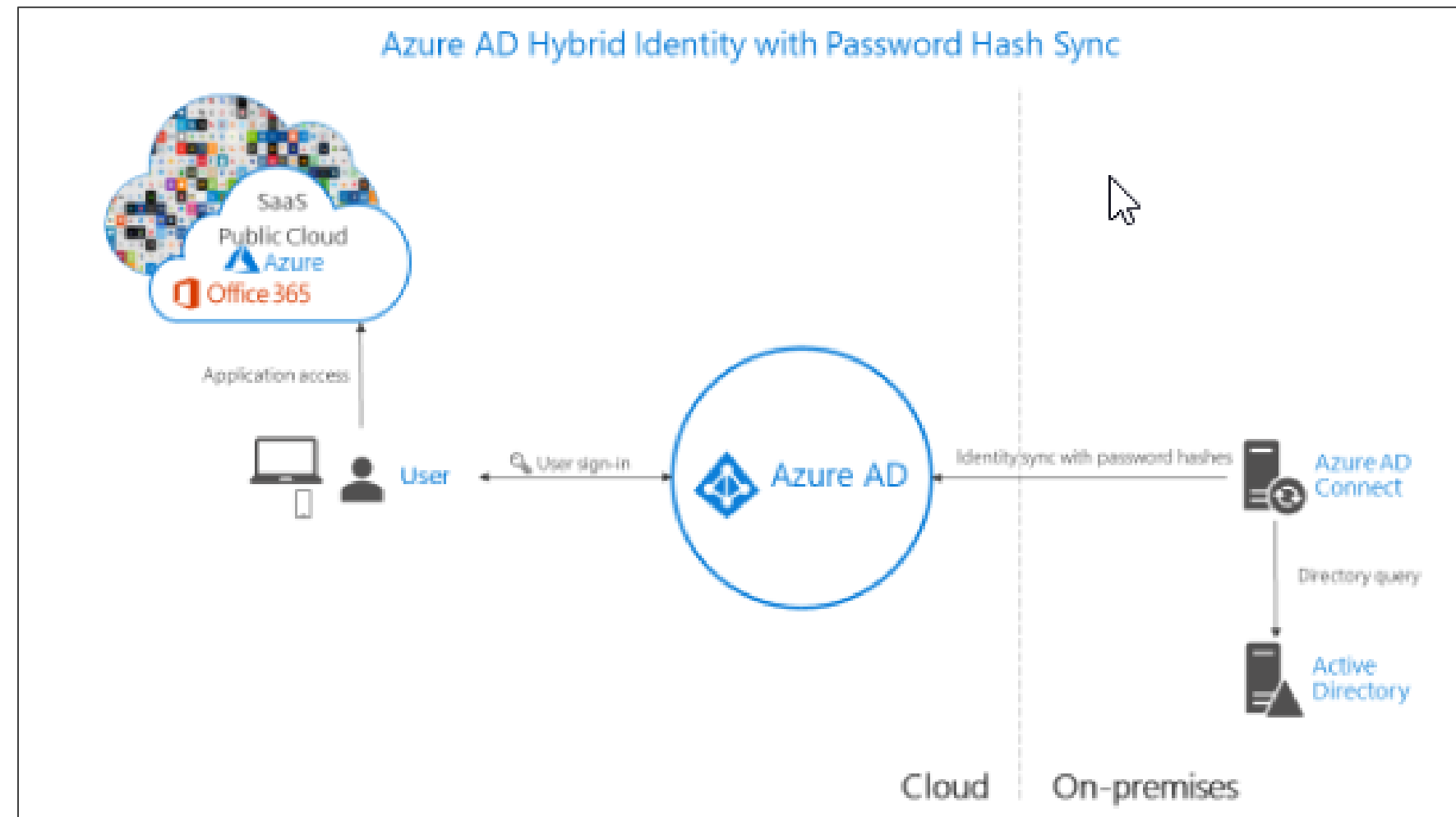


Source: <https://learn.microsoft.com/en-us/entra/identity/hybrid/connect/choose-ad-authn>

Authentication Architecture

The following diagrams outline the high-level architecture components required for each authentication method

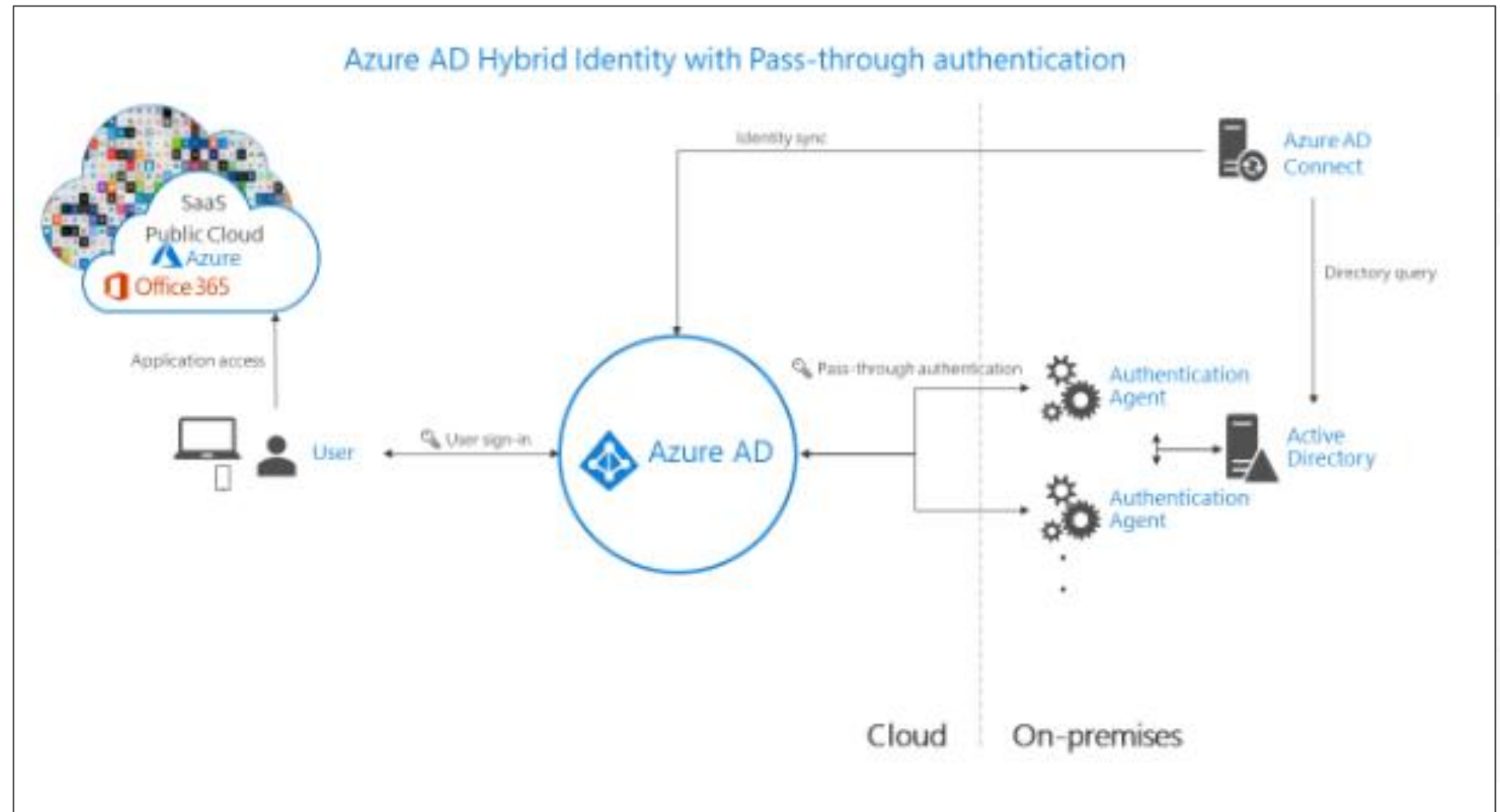
Simplicity of a password hash synchronization solution:



Source: <https://learn.microsoft.com/en-us/entra/identity/hybrid/connect/choose-ad-authn>

Authentication Architecture

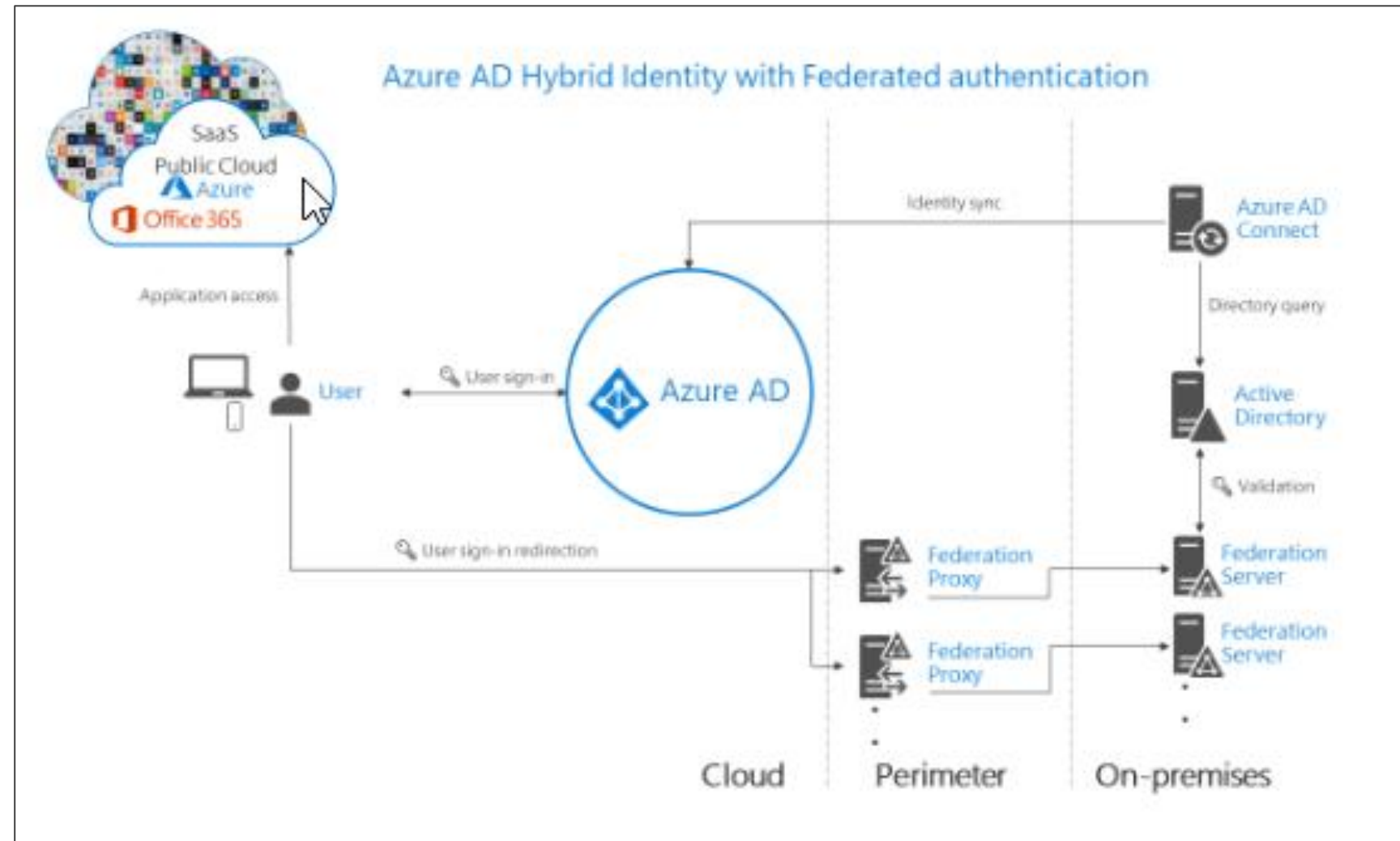
Agent requirements of pass-through authentication using two agents for redundancy:



Source: <https://learn.microsoft.com/en-us/entra/identity/hybrid/connect/choose-ad-authn>

Authentication Architecture

Components required for federation in the perimeter and internal network of the organization:

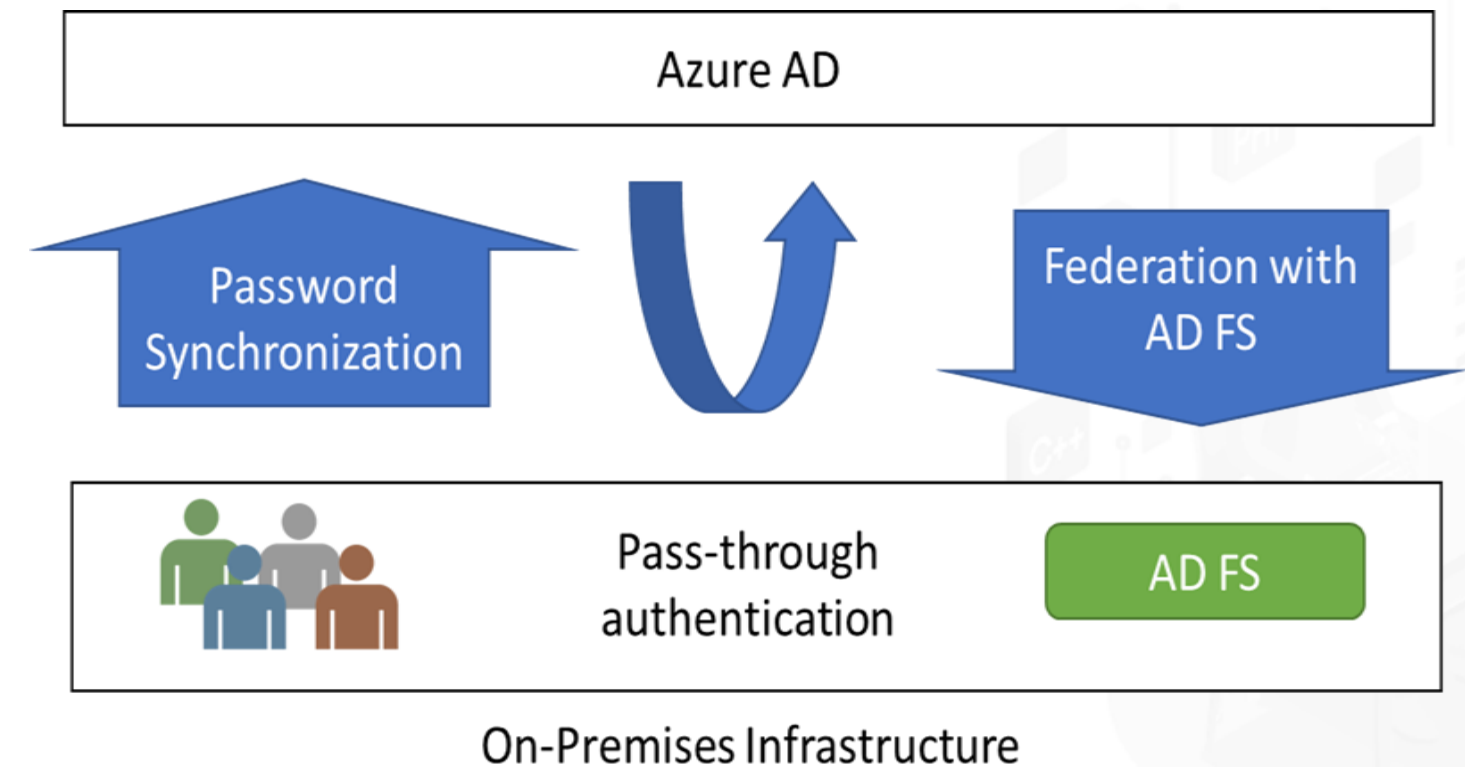


Authentication Options

The first option organizations can choose to authenticate is:

Password Hash Synchronization (PHS)

PHS can synchronize an encrypted version of the password hash for user accounts.

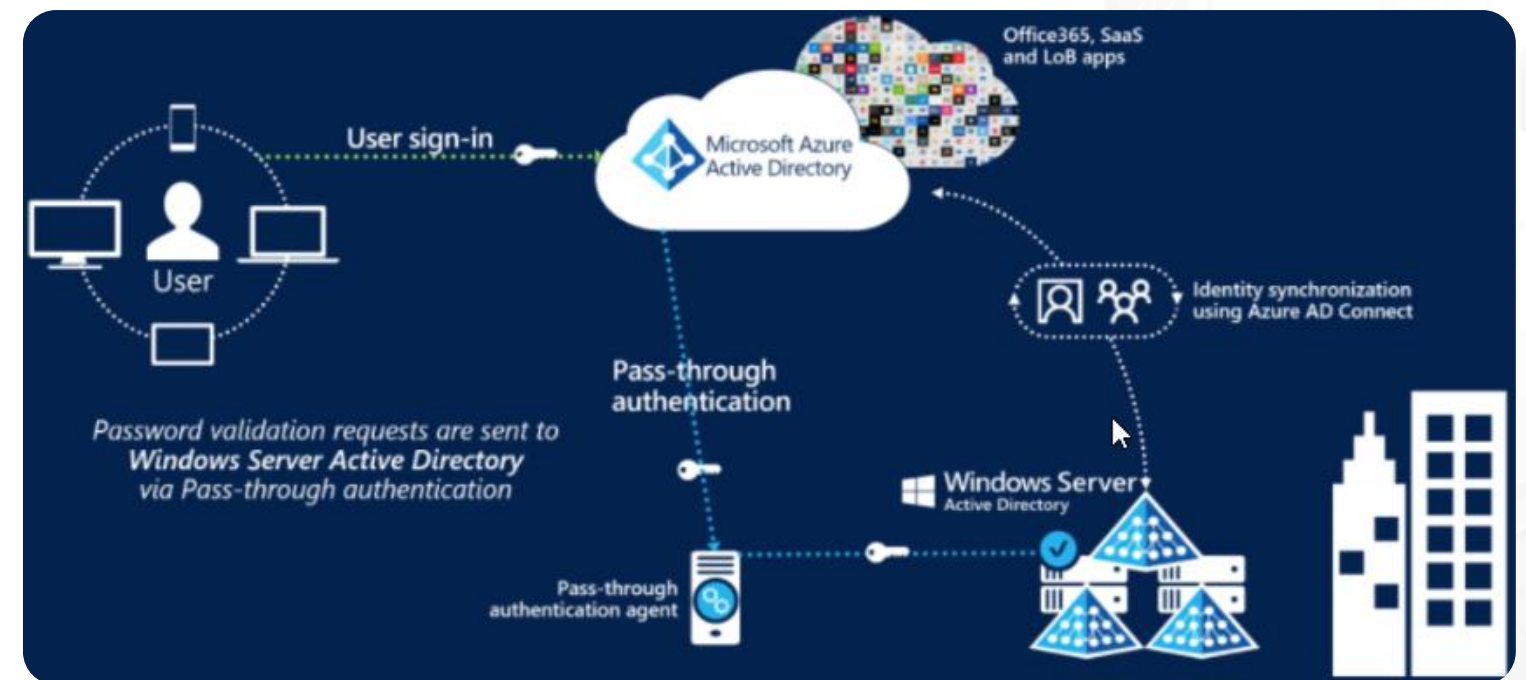


Authentication Options

The second option organizations can choose to authenticate is:

Pass-through Authentication (PTA)

It authenticates the username and password with the on-premises domain controllers.

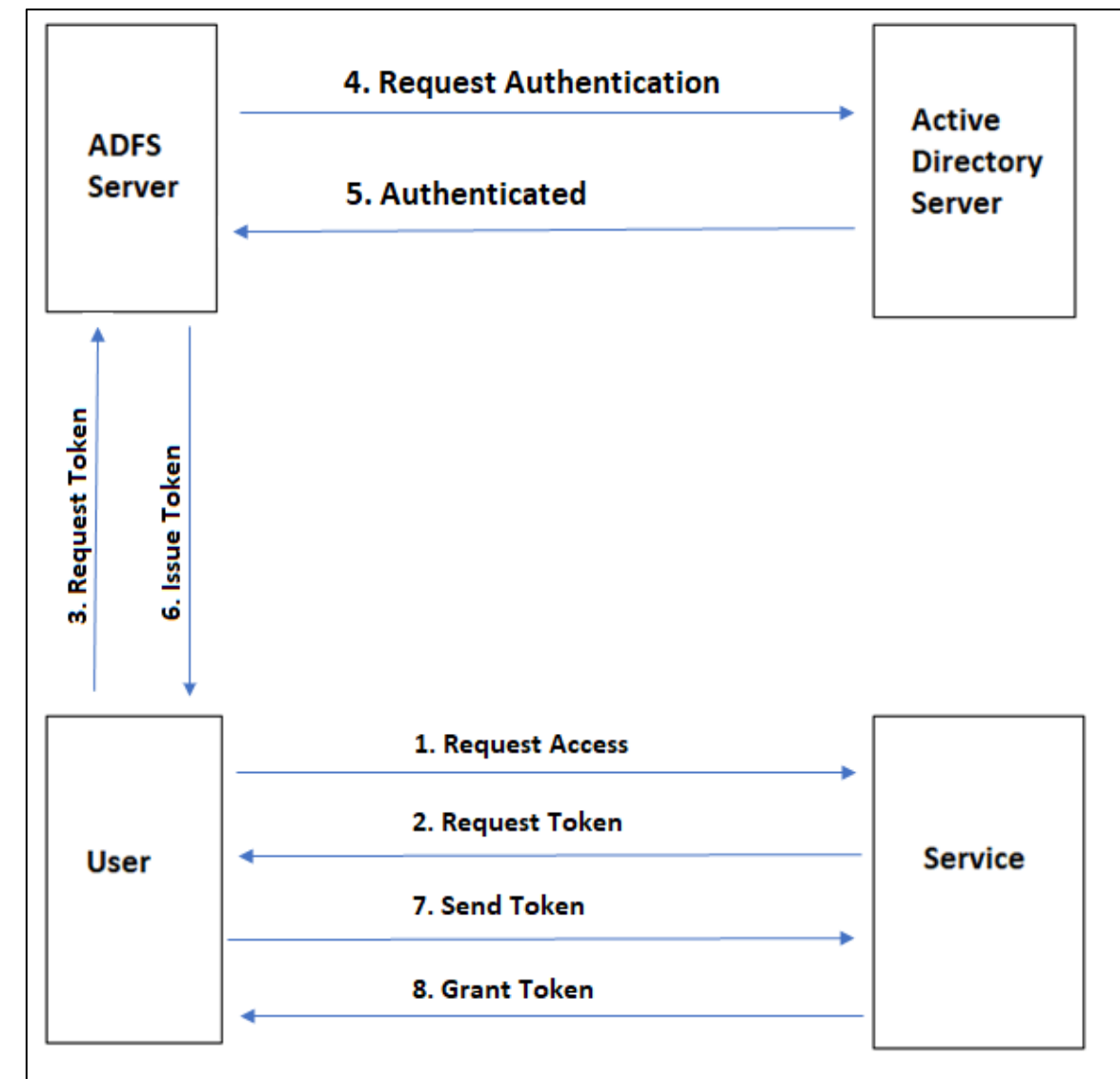


Authentication Options

The third option organizations can choose to authenticate is:

Active Directory Federation Services

AD FS is the Microsoft implementation of an identity federation solution that uses claims-based authentication.



Comparing Authentication Methods

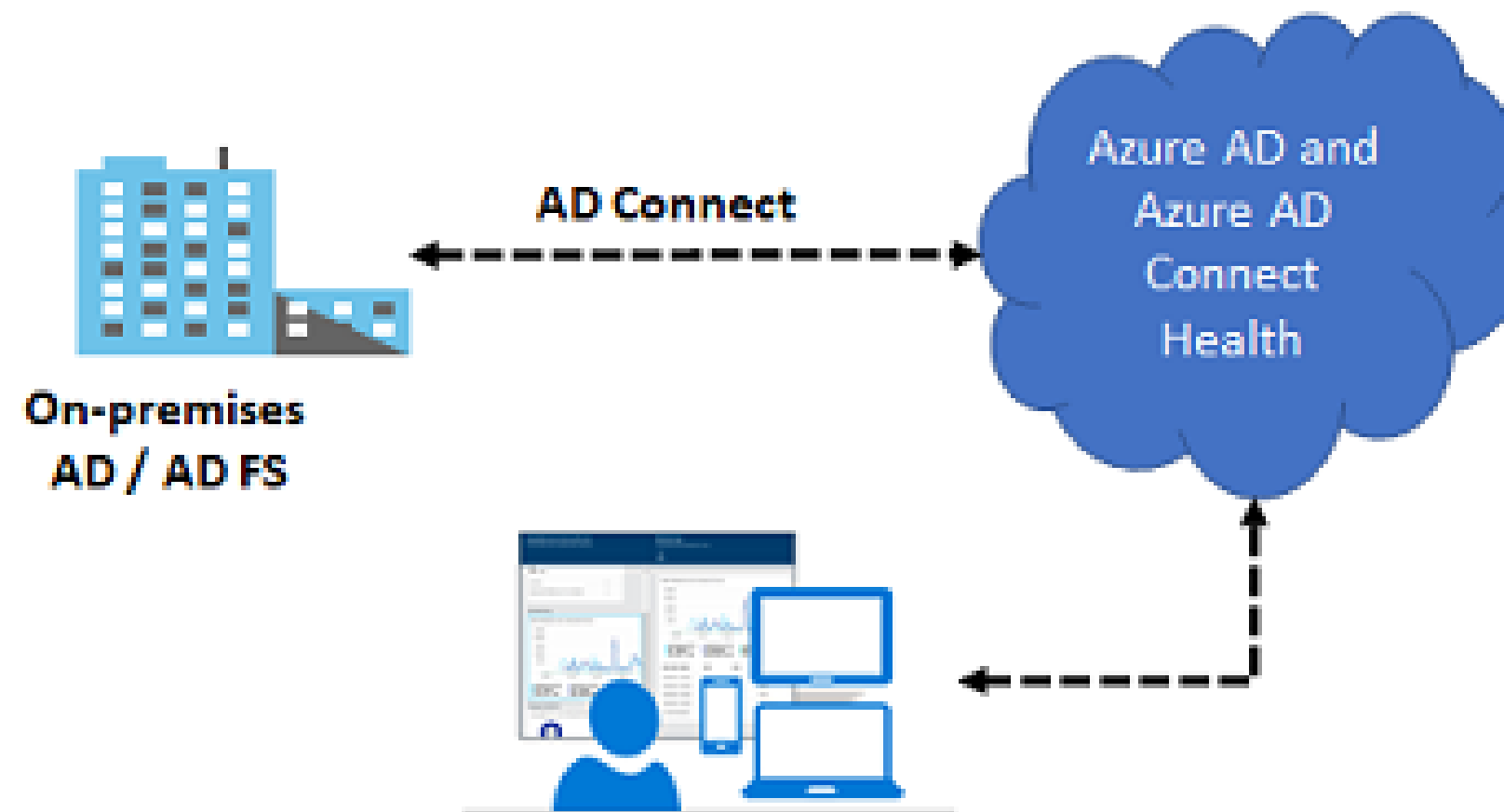
Consideration	Password hash synchronization + Seamless SSO	Pass-through Authentication + Seamless SSO
Where does authentication happen?	In the cloud	In the cloud, after a secure password verification exchange with the on-premises authentication agent
What are the on-premises server requirements beyond the provisioning system: Azure AD Connect?	None	One server for each additional authentication agent
What are the requirements for on-premises internet and networking beyond the provisioning system?	None	Outbound internet access from the servers running authentication agents
Is there a TLS/SSL certificate requirement?	No	No
Is there a health monitoring solution?	Not required	Microsoft Entra admin center provides agent status

Comparing Authentication Methods

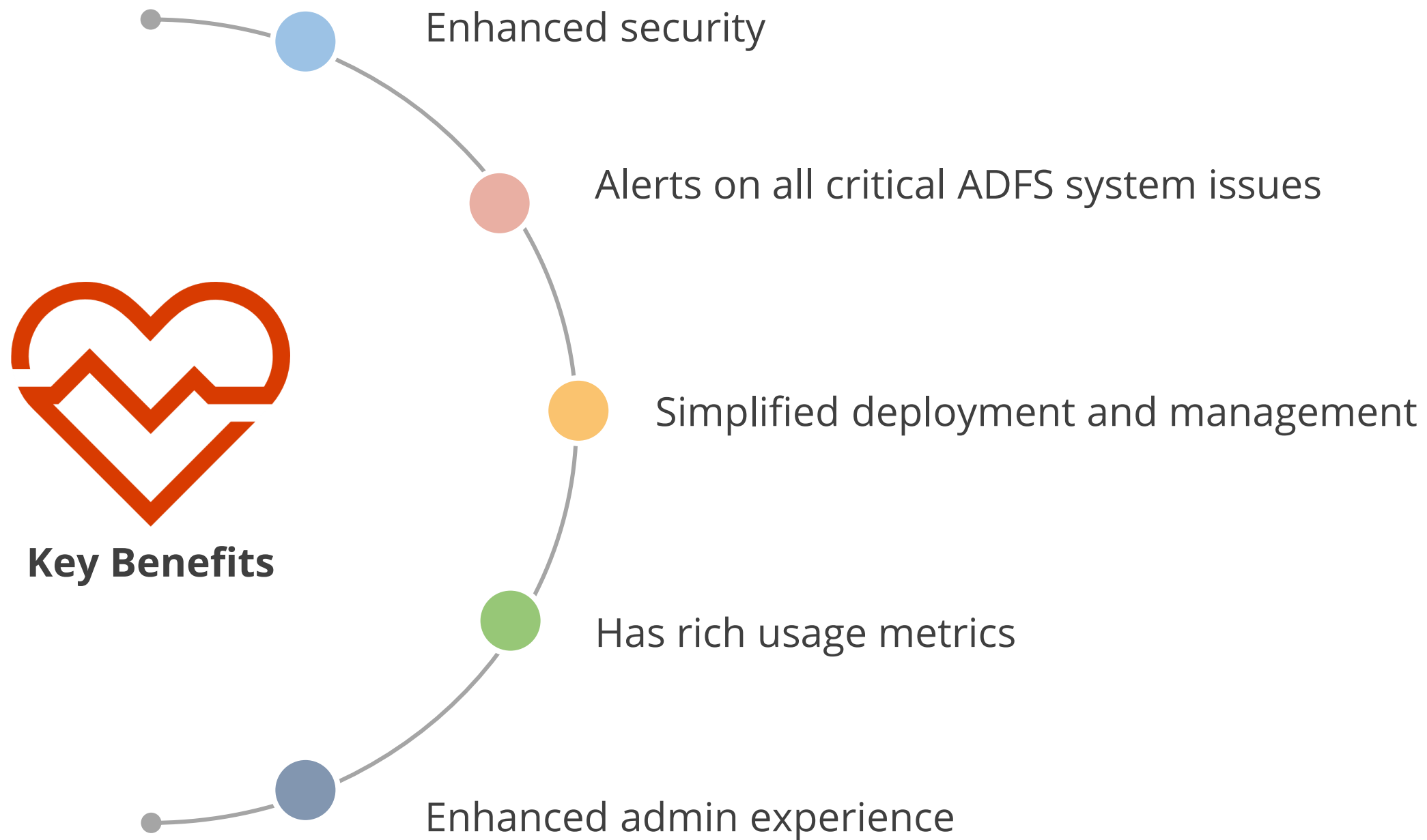
Consideration	Password hash synchronization + Seamless SSO	Pass-through authentication + Seamless SSO
Do users get a single sign-on to cloud resources from domain-joined devices within the company network?	Yes, with seamless SSO	Yes, with seamless SSO
What sign-in types are supported?	UserPrincipalName + password Windows-Integrated authentication by using seamless SSO alternate login ID	UserPrincipalName + password Windows-Integrated authentication by using seamless SSO alternate login ID
What are the multi-factor authentication options?	Microsoft Entra MFA Custom controls with conditional access	Microsoft Entra MFA Custom controls with conditional access
What are the conditional access options?	Microsoft Entra conditional access with Microsoft Entra ID P1 or P2	Microsoft Entra conditional access with Microsoft Entra ID P1 or P2

Microsoft Entra ID Connect Health

It provides robust monitoring of on-premises identity infrastructure. Thus, it ensures the reliability of the environment.

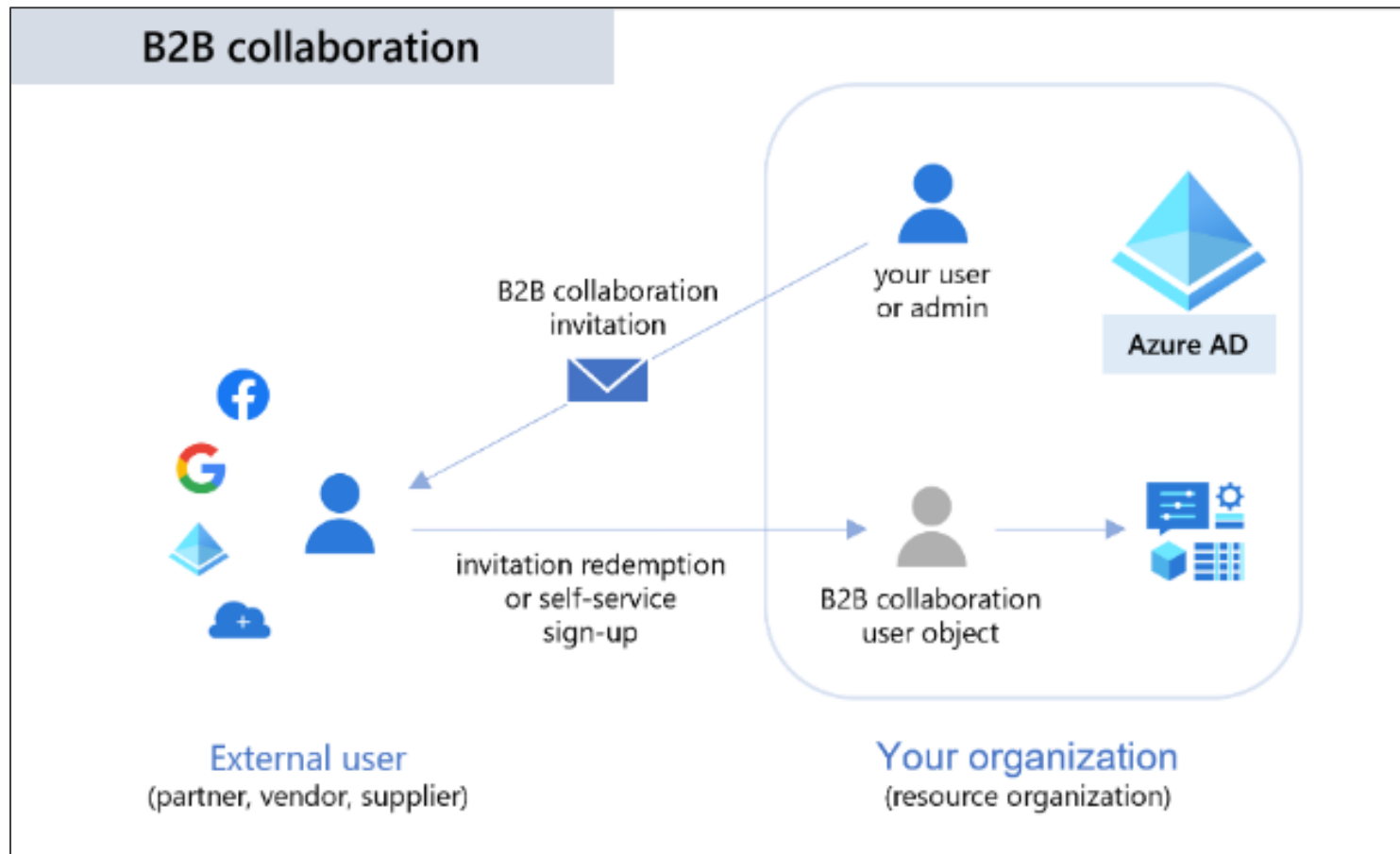


Why Use Microsoft Entra ID Connect Health?



Implement a Solution for B2B Integration

Microsoft Entra ID B2B



- There are no external operating costs for the user's business.
- The partner uses their own identities and credentials.
- The user need not manage external accounts and passwords.
- The user need not sync accounts or manage account lifecycles.

Guest Users

These are the users who are not considered an internal entity, such as an external partner, stakeholder, or customer.

Prerequisites

- Anyone can be asked to work for a company by adding them as a guest user.
- Guest users can use their own work, education, or social identity to log in.
- A user should have the ability to create user accounts.
- A user should have a working email address.

Adding Guest Users

Anyone can work with a company by being added to a directory as a guest user.



Follow the instructions given below to add a new guest user:

- Log in as an administrator to the Microsoft Azure portal
- Create a new guest user
- The user will receive the invitation as a guest
- After accepting the invite, guest users can be assigned to any app or group

Adding Guest Users

Allow application or group owners to manage their guest users.

1

Administrators:
Self-service app
and group
management

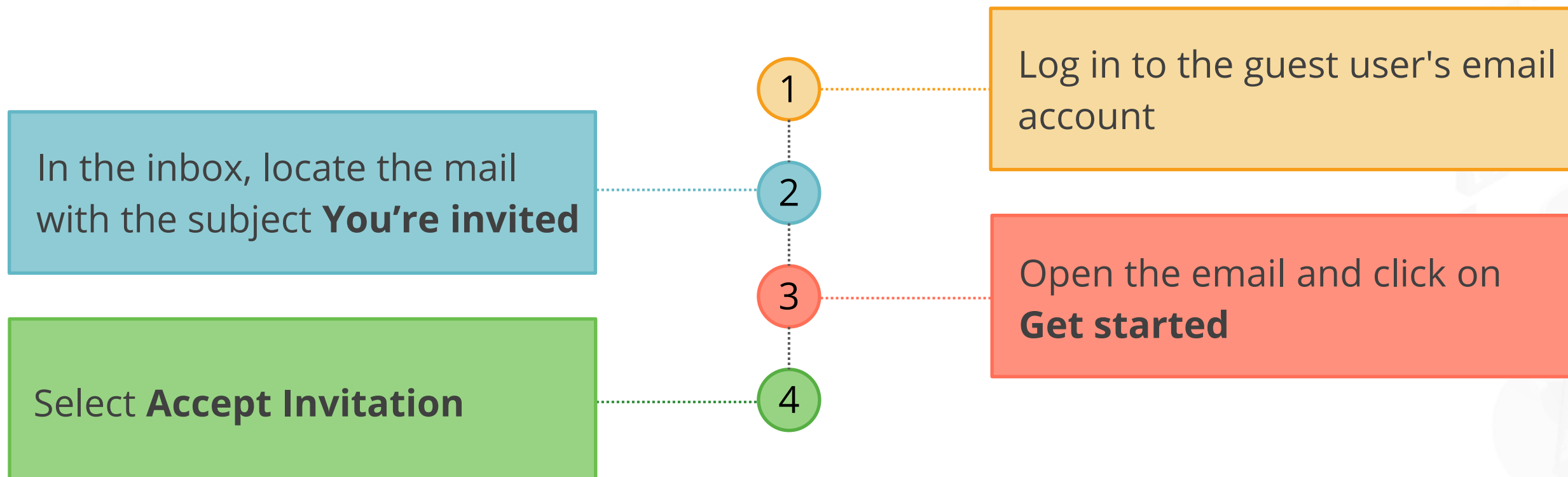
2

**Non-
administrators:**
Use access panel to
add guest users



Accept the Guest User Invite

Follow the instructions to accept a new user invite:



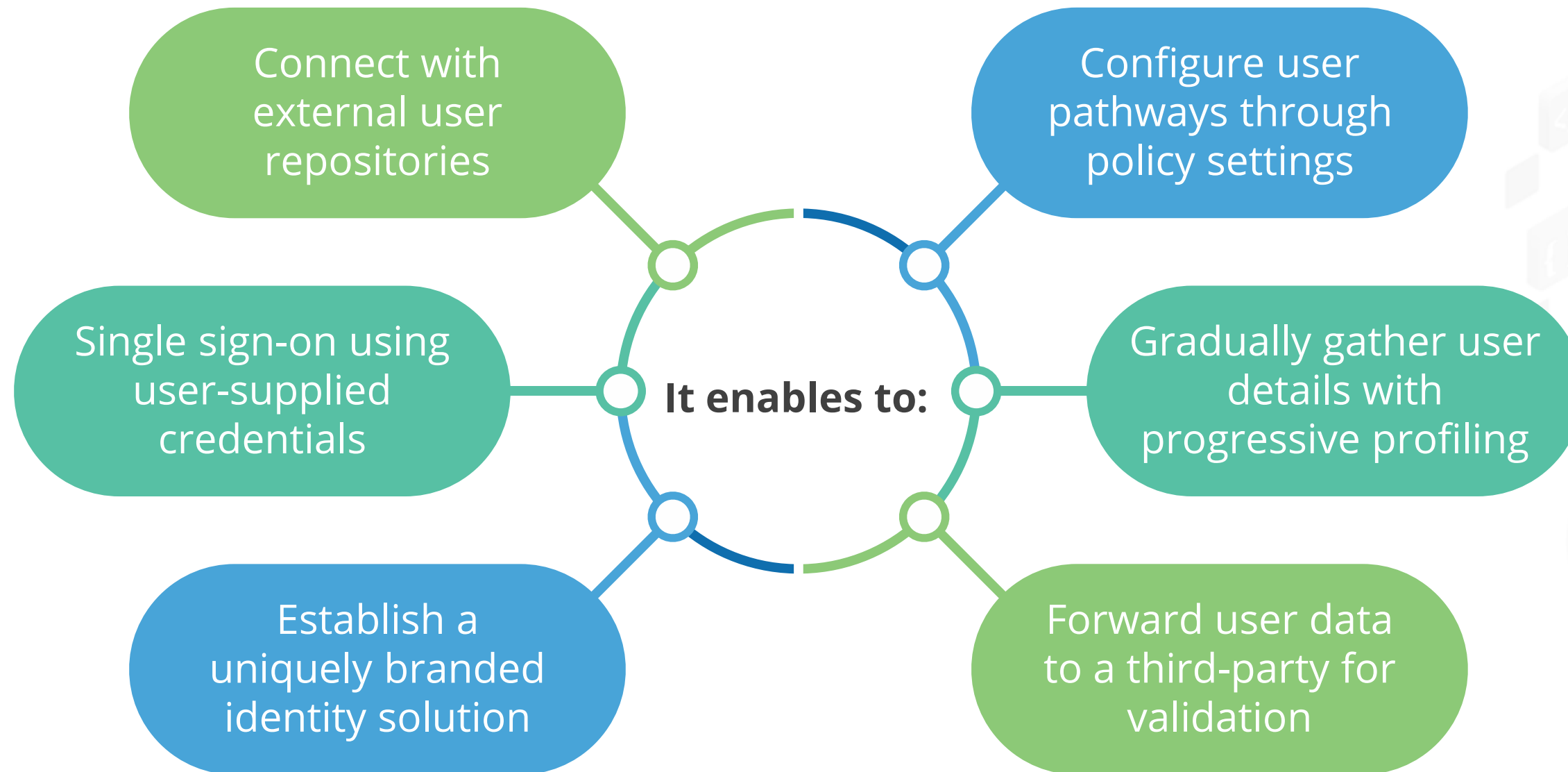
Azure AD B2C

Azure AD B2C

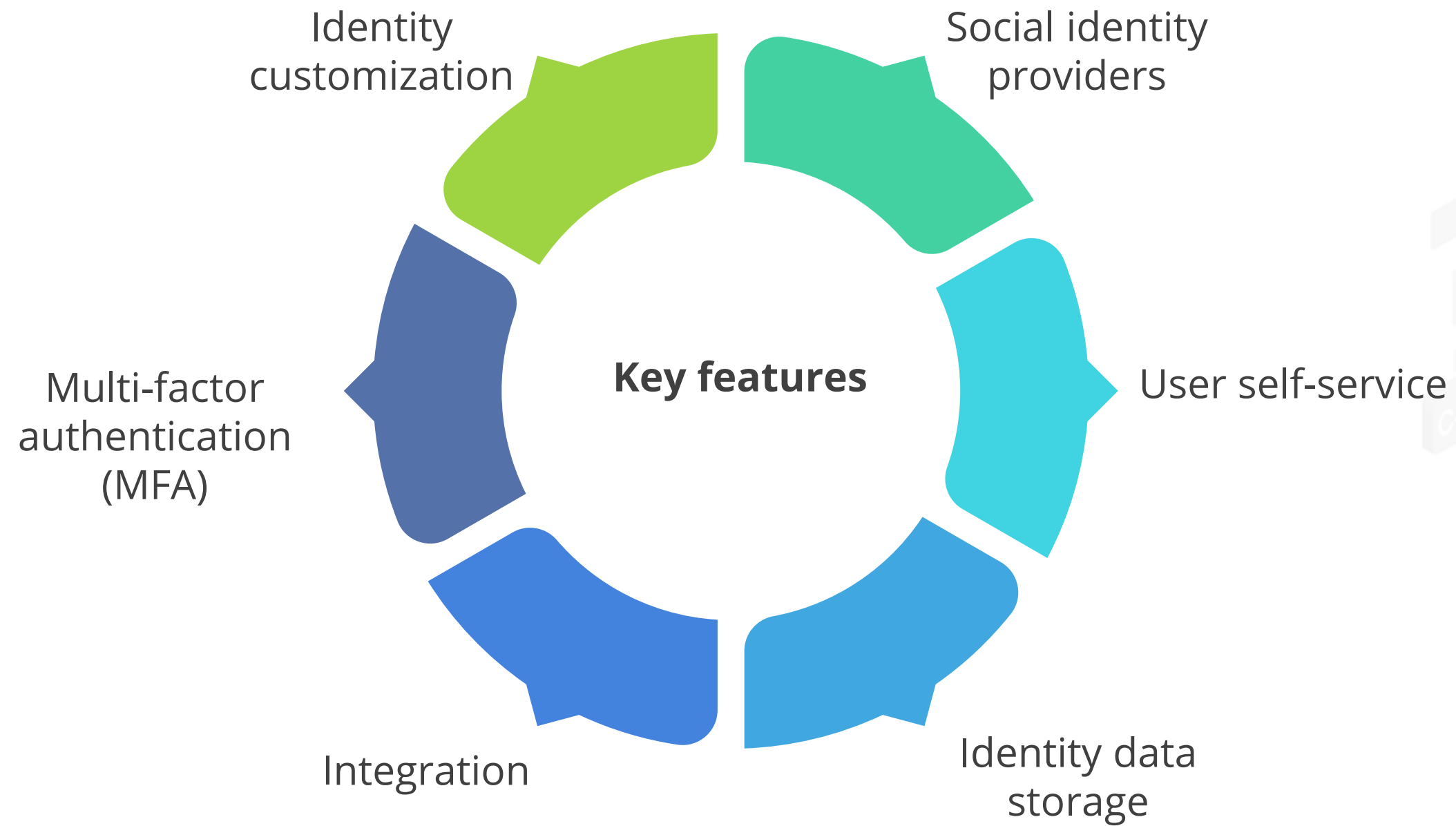


- It is a cloud-based identity and access management service provided by Microsoft.
- It is designed to enable organizations to manage customer-facing applications.

Azure AD B2C



Azure AD B2C: Features



Azure AD B2C

Use cases are as follows:



Creating Microsoft Entra Join



Duration: 10 Min.

Problem Statement:

You've asked an Azure Architect to assist your company with an Azure authentication solution that allows access to both cloud and on-premises apps and resources.

Outcome:

You will be able to create an Azure authentication solution that provides seamless access to both cloud and on-premises applications and resources.

Note: Refer to the demo document for detailed steps:
[02_Creating_Microsoft_Entra_Join](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Create a Microsoft Entra ID join
2. Create a virtual machine
3. Connect RDP (Remote Desktop Protocol) to the virtual machine to join the machine to the Microsoft Entra ID domain



Creating Microsoft Entra ID User



Duration: 10 Min.

Problem Statement:

As an Azure Architect, you have been asked to help your organization with an Azure authentication solution that can help manage the users.

Outcome:

You will be able to create an Azure authentication solution that effectively manages users, providing secure and seamless access to organizational resources.

Note: Refer to the demo document for detailed steps:
[03_Creating_Microsoft_Entra_ID_User](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Create a user in Microsoft Entra ID



Creating Microsoft Entra ID Guest User



Duration: 10 Min.

Problem Statement:

As an Azure Architect, you've been asked to assist your company with an Azure authentication solution that will allow external users to collaborate with your company.

Outcome:

You will be able to create an Azure authentication solution that allows external users to securely collaborate with your company.

Note: Refer to the demo document for detailed steps:
04_Creating_Microsoft_Entra_ID_Guest_User

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Create a guest user in Microsoft Entra ID



Creating Microsoft Entra ID Group Creation



Duration: 10 Min.

Problem Statement:

As an Azure Architect, you've been asked to provide your company with an Azure authentication solution for managing members and computer access to shared resources for a group of users.

Outcome:

You will be able to create and manage Azure authentication solutions for groups of users, enabling secure access to shared resources and efficient member management.

Note: Refer to the demo document for detailed steps:
05_Creating_Microsoft_Entra_ID_Group_Creation

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Create a user group in Microsoft Entra ID



Adding Microsoft Entra ID Authentication for Storage



Duration: 10 Min.

Problem Statement:

As an Azure Architect, you have been asked to help your organization with an Azure authentication solution to authorize requests to Blob and Queue storage.

Outcome:

You will be able to configure Azure authentication solutions to securely authorize requests to Blob and Queue storage, ensuring seamless access control for your organization's storage needs.

Note: Refer to the demo document for detailed steps:
[06_Adding_Microsoft_Entra_ID_Authentication_for_Storage](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed:

1. Create a new user in Microsoft Entra ID
2. Create Azure storage account and upload blob to the storage account
3. Download Azure Storage Explorer
4. Add Microsoft Entra ID authentication for storage



Adding Microsoft Entra ID Authentication for Storage



Duration: 10 Min.

Problem Statement:

As an Azure Architect, you have been asked to help your organization with an Azure authentication solution to create a custom domain name that will help you to create usernames familiar to your users.

Outcome:

You will be able to set up a custom domain name within Azure, enabling your organization to use familiar and user-friendly usernames for authentication purposes.

Note: Refer to the demo document for detailed steps:
[07_Adding_Microsoft_Entra_ID_Authentication_for_Storage](#)

ASSISTED PRACTICE

Unassisted Practice: Guidelines

Steps to be followed:

1. Create a custom domain in Microsoft Entra ID



Key Takeaways

- In conditional access policies, if users want to access a resource, then they must complete an action.
- Multi-factor authentication is a process where a user is prompted during the sign-in process for an additional form of identification.
- Microsoft Entra ID seamless SSO automatically signs users in when they are on their corporate devices connected to a corporate network.
- Guest users are not expected to receive an internal invitation from the CEO or to receive any company benefits.





Knowledge Check

What is the purpose of Microsoft Entra ID Connect Health?

- A. To monitor the health and performance of Azure virtual machines
- B. To manage access control and permissions in Microsoft Entra ID
- C. To synchronize on-premises directories with Microsoft Entra ID
- D. To monitor the health and performance of on-premises identity infrastructure



What is the purpose of Microsoft Entra ID Connect Health?

- A. To monitor the health and performance of Azure virtual machines
- B. To manage access control and permissions in Microsoft Entra ID
- C. To synchronize on-premises directories with Microsoft Entra ID
- D. To monitor the health and performance of on-premises identity infrastructure



The correct answer is **D**

Azure AD Connect Health is a feature that helps monitor the health and performance of on-premises identity infrastructure.

**Knowledge
Check**
02

Which authentication method in Microsoft Entra ID is suitable for environments that require users to enter their on-premises credentials for authentication?

- A. Password Hash Synchronization (PHS)
- B. Pass-through Authentication (PTA)
- C. Active Directory Federation Services (AD FS)
- D. Azure Multi-Factor Authentication (MFA)



Knowledge
Check
02

Which authentication method in Microsoft Entra ID is suitable for environments that require users to enter their on-premises credentials for authentication?

- A. Password Hash Synchronization (PHS)
- B. Pass-through Authentication (PTA)
- C. Active Directory Federation Services (AD FS)
- D. Azure Multi-Factor Authentication (MFA)



The correct answer is **C**

Active Directory Federation Services (AD FS) is suitable for environments where users need to enter their on-premises credentials for authentication.

What is the purpose of Microsoft Entra ID self-service sign-up?

- A. To enable automatic sign-up and provisioning of Microsoft Entra ID accounts for new employees
- B. To allow external users to create their own Microsoft Entra ID accounts to access apps and services
- C. To simplify the user onboarding process by automating user creation in Microsoft Entra ID
- D. To provide a single sign-on experience for users across on-premises and cloud-based applications



Knowledge Check

03

What is the purpose of Microsoft Entra ID self-service sign-up?

- A. To enable automatic sign-up and provisioning of Microsoft Entra ID accounts for new employees
- B. To allow external users to create their own Microsoft Entra ID accounts to access apps and services
- C. To simplify the user onboarding process by automating user creation in Microsoft Entra ID
- D. To provide a single sign-on experience for users across on-premises and cloud-based applications



The correct answer is **B**

Azure AD self-service sign-up allows external users to create their own Azure AD accounts.

Which authentication method is recommended for organizations that want to keep the on-premises password hashes out of the cloud?

- A. Password Hash Synchronization (PHS)
- B. Pass-through Authentication (PTA)
- C. Active Directory Federation Services (AD FS)
- D. Azure Multi-Factor Authentication (MFA)



Knowledge
Check

04

Which authentication method is recommended for organizations that want to keep the on-premises password hashes out of the cloud?

- A. Password Hash Synchronization (PHS)
- B. Pass-through Authentication (PTA)
- C. Active Directory Federation Services (AD FS)
- D. Azure Multi-Factor Authentication (MFA)



The correct answer is **B**

Pass-through authentication allows users to sign in to Microsoft Entra ID and other Microsoft 365 services using the same password they use on-premises.